SAP ID - 590042185

NAME – Piyush Negi

EXPERIMENT – 1

Linux Programming Environment & Basic Commands

A) Setting up of Linux Programming Environment :

We can do C programming online or offline according to the needs. It is required for us to keep working on a Linux Environment.

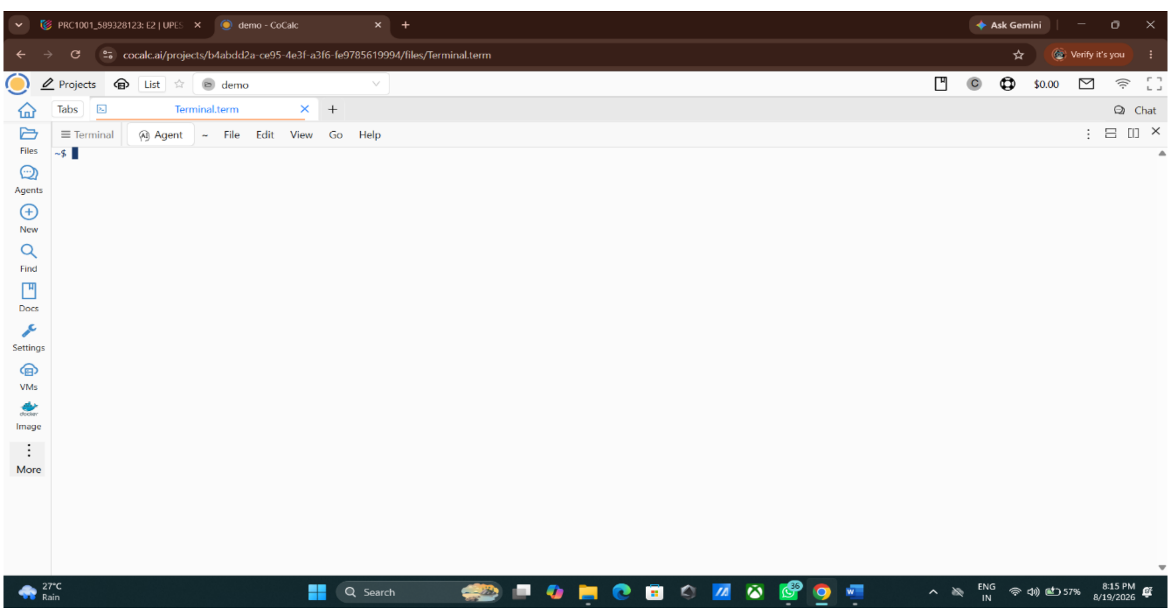
- Online Method – Cocalc : CoCalc is an online website that provides a linux based programming environment using web browser.

Steps:

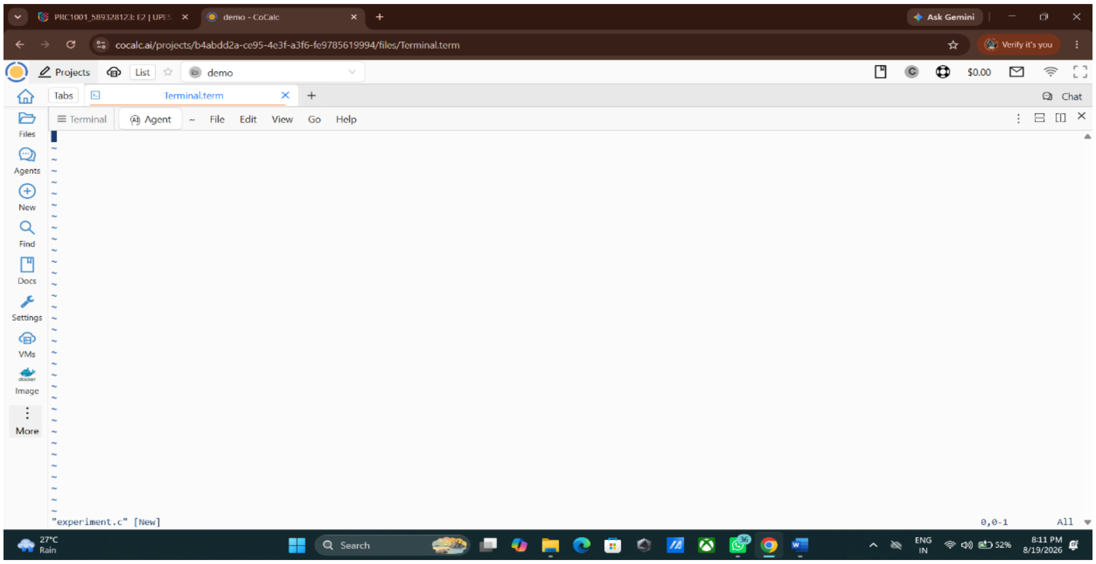
1) Open the CoCalc website and create or login to an account.

2)Create a new project.

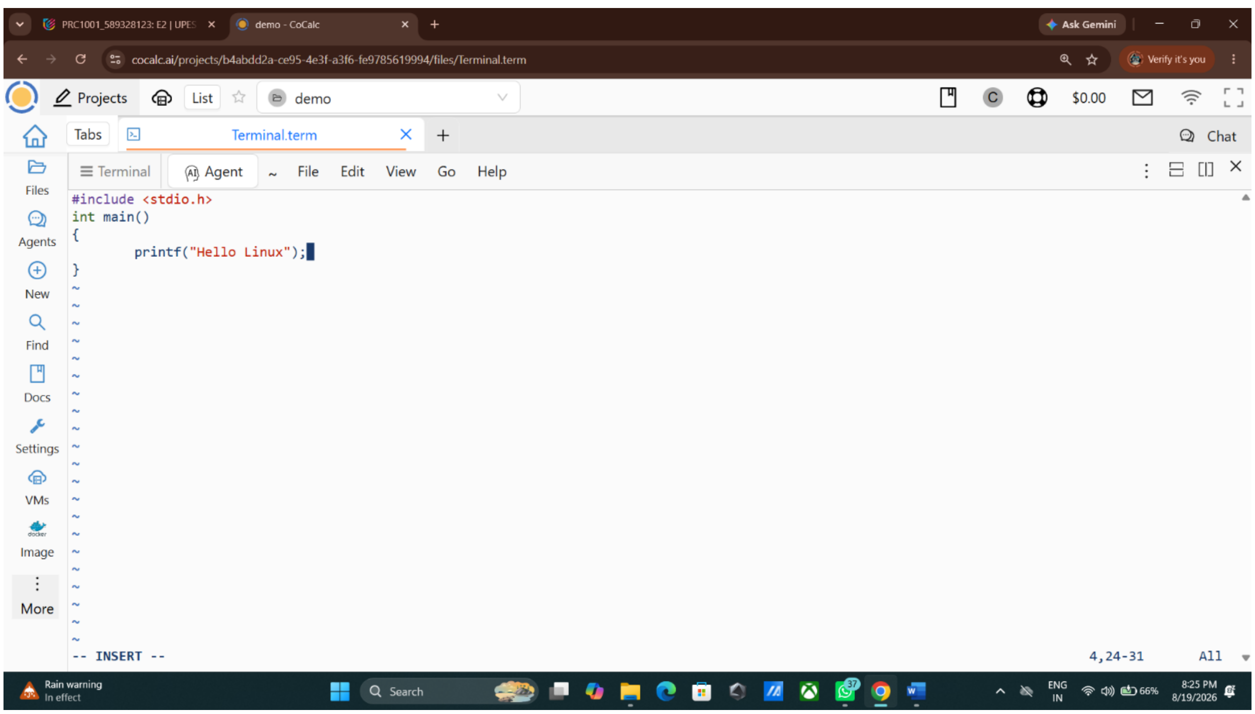
3) Open the **Linux Terminal** provided in the project.



4) Create a C source file using a text editor.



5) Write the C program and save the file.

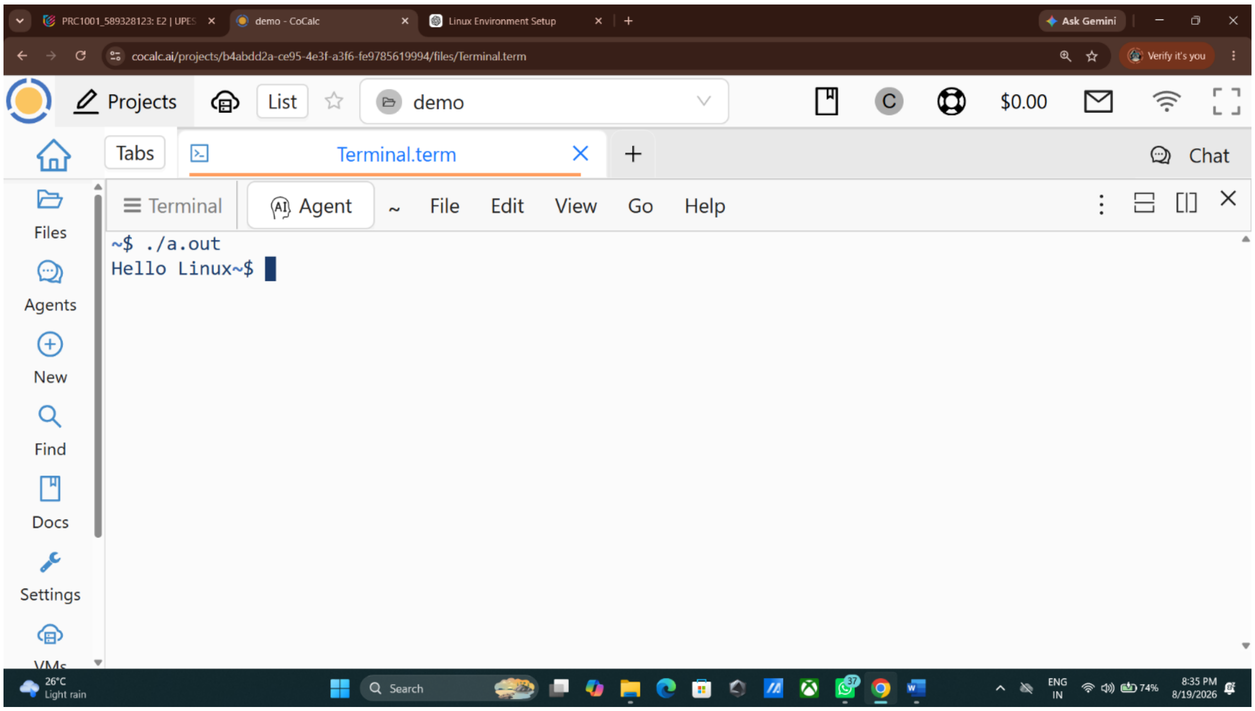


6) Compile the program using:

cc experiment.c

7) Execute it using :

./a.out



- Offline Methods – Linux can used on a computer through some methods.

  1) Dual Boot : Linux and other operating systems such as Windows are installed on the same computer.

  2) Single Boot: Linux is installed as the only operating system on the computer.

  3) Bootable Pendrive: Linux is run from a bootable USB drive. It can be used to install Linux on the computer.

  4) **Virtual Machine:** Linux is installed inside a virtual machine that runs on an existing operating system. Through this you can run linux without changing the operating system.

Virtual Machine Manager (Hypervisor)

A hypervisor is software or firmware that creates and manages virtual machines (VMs). It allows multiple operating systems to run on the same physical computer.

Type-1 Hypervisor – Bare Metal
Type-1 hypervisor runs directly on the physical hardware without requiring a conventional host operating system.
Examples: VMware ESXi , Microsoft Hyper-V, Xen etc.

Advantages: Better performance and efficient use of hardware resources.

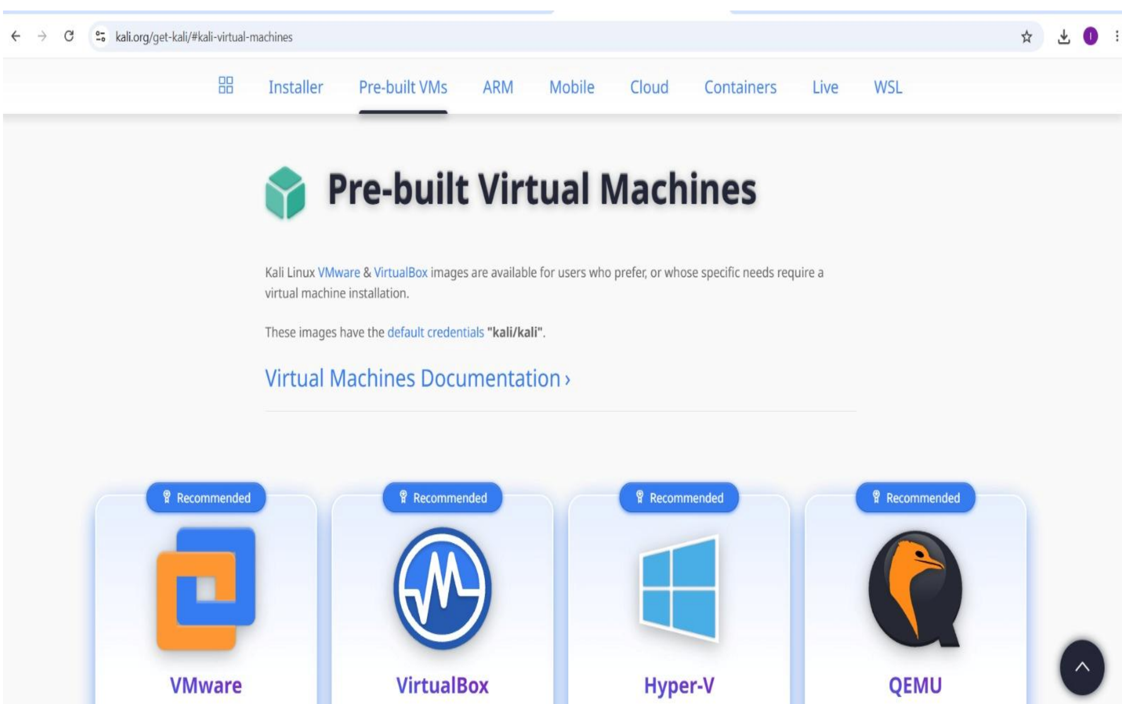
Type-2 Hypervisor – Hosted
Type-2 hypervisor runs as an application on top of an existing host operating system. The virtual machines are then created and managed through this software.
Example: Oracle VirtualBox, VMware Workstation, VMware Fusion etc.

Advantages: Easy to install and use, especially for learning and development.

2) **Installation of VMM & VM**

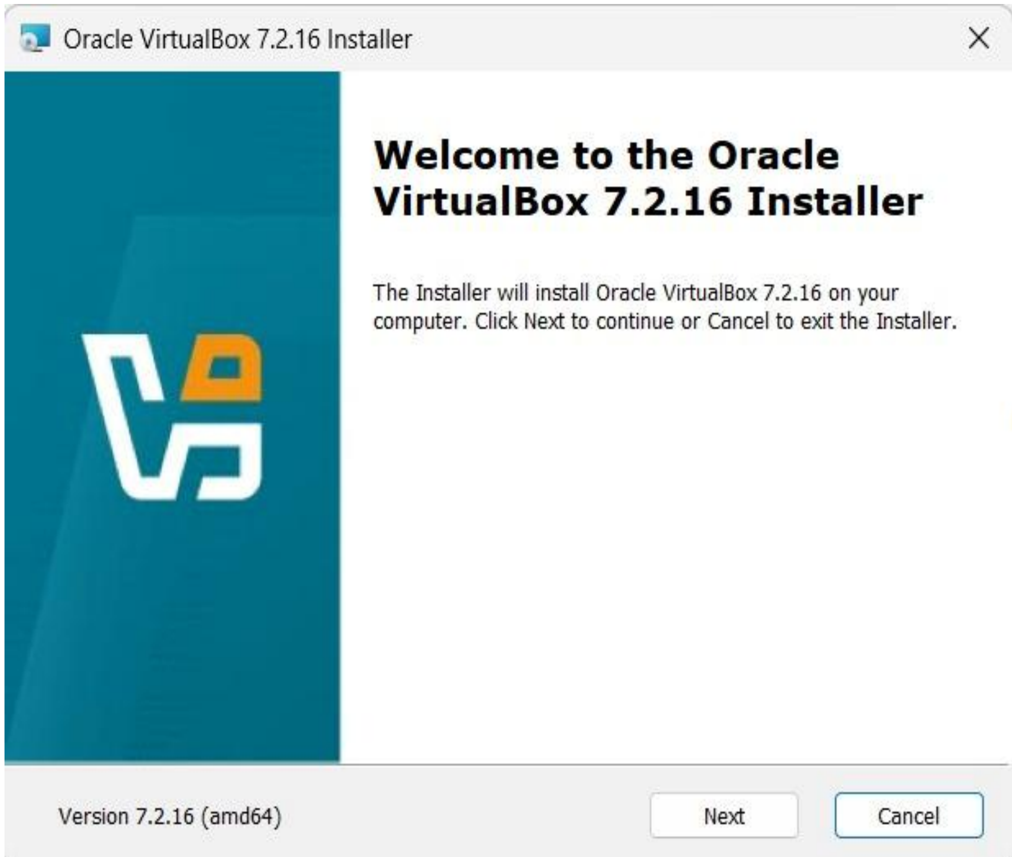
Step-1: Download Oracle Virtual Box for “Windows hosts”

Step-2: Download Pre-built Kali Linux VM for Virtualbox.

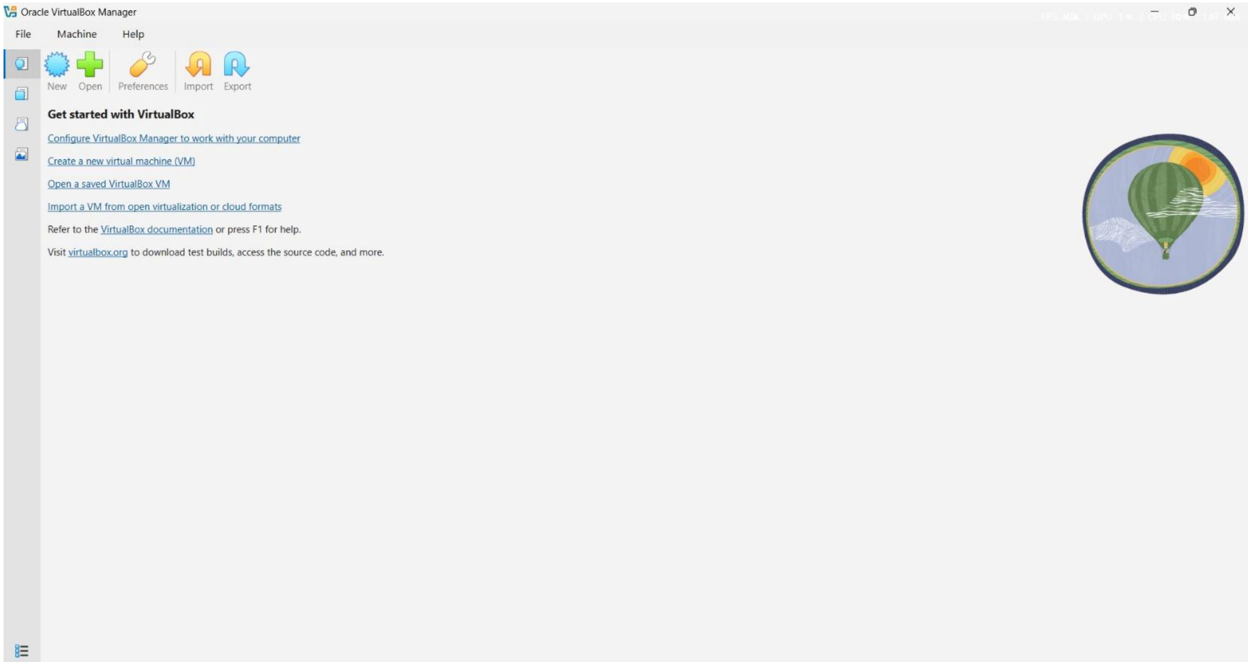


Step-3: Extracted the Kali Linux downloaded file

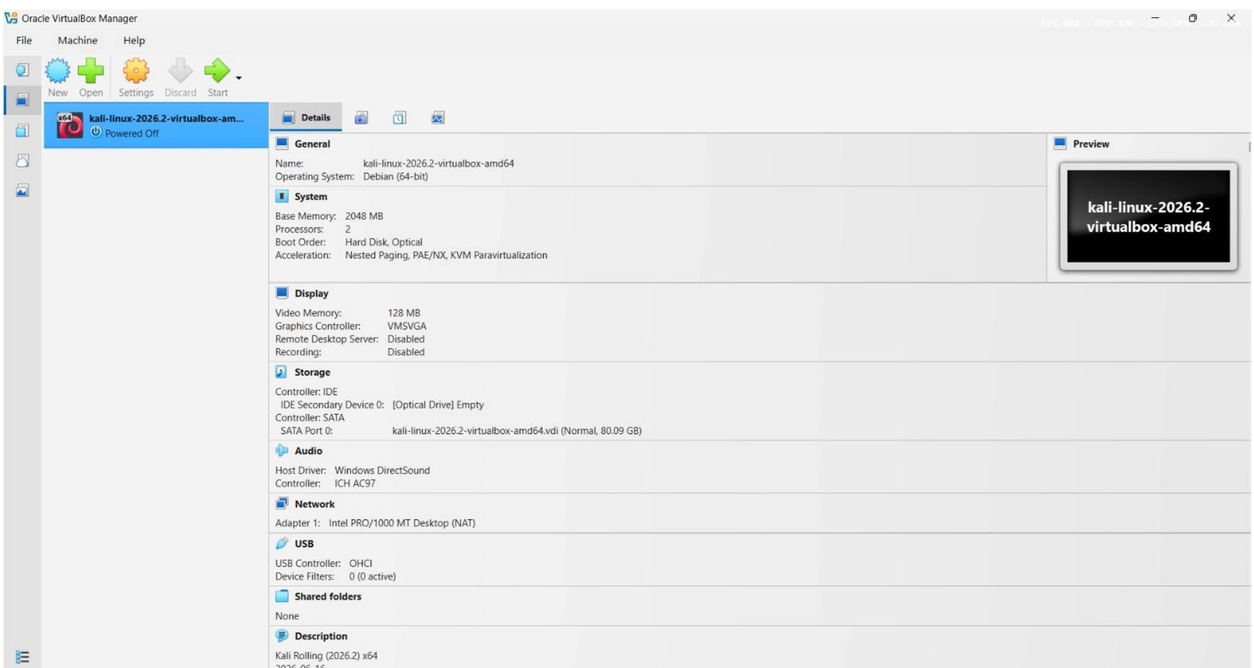
Step-4: Install Virtualbox



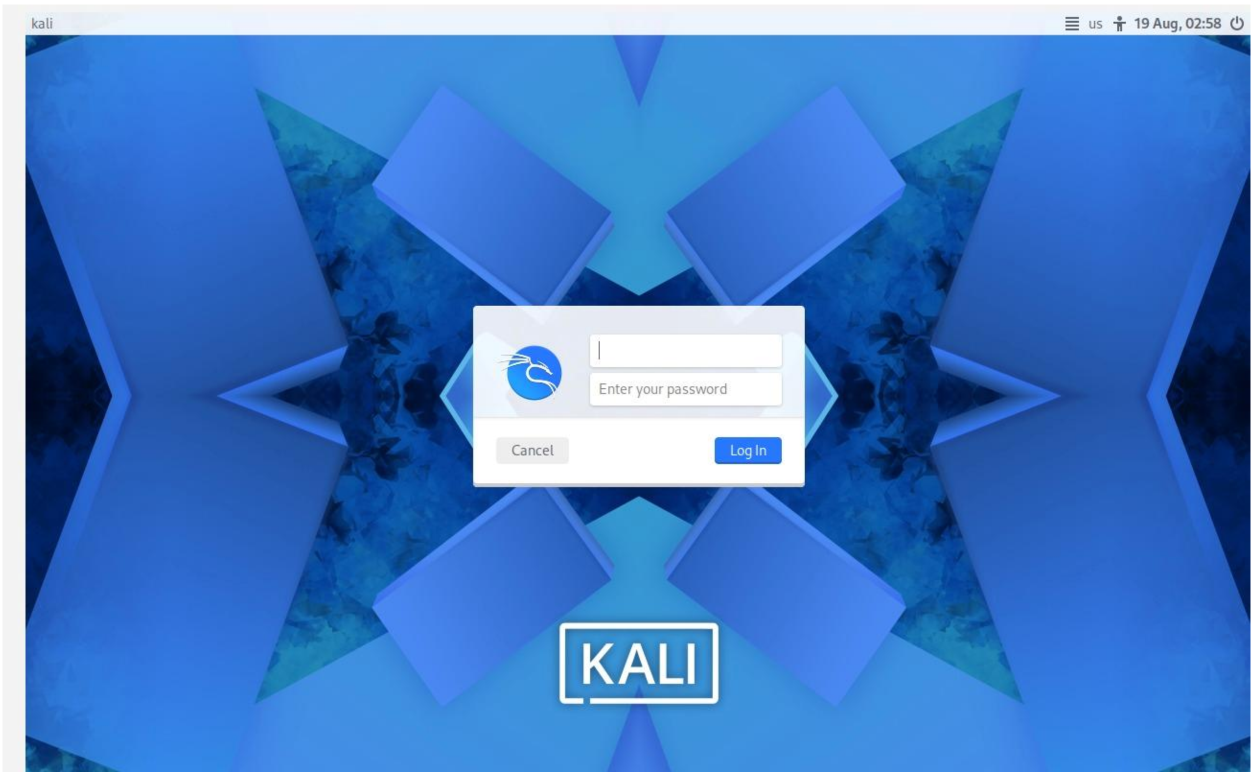
Step-5: Open Virtual box



Step-6: Open the “saved VM”



Step-7: Start the VM



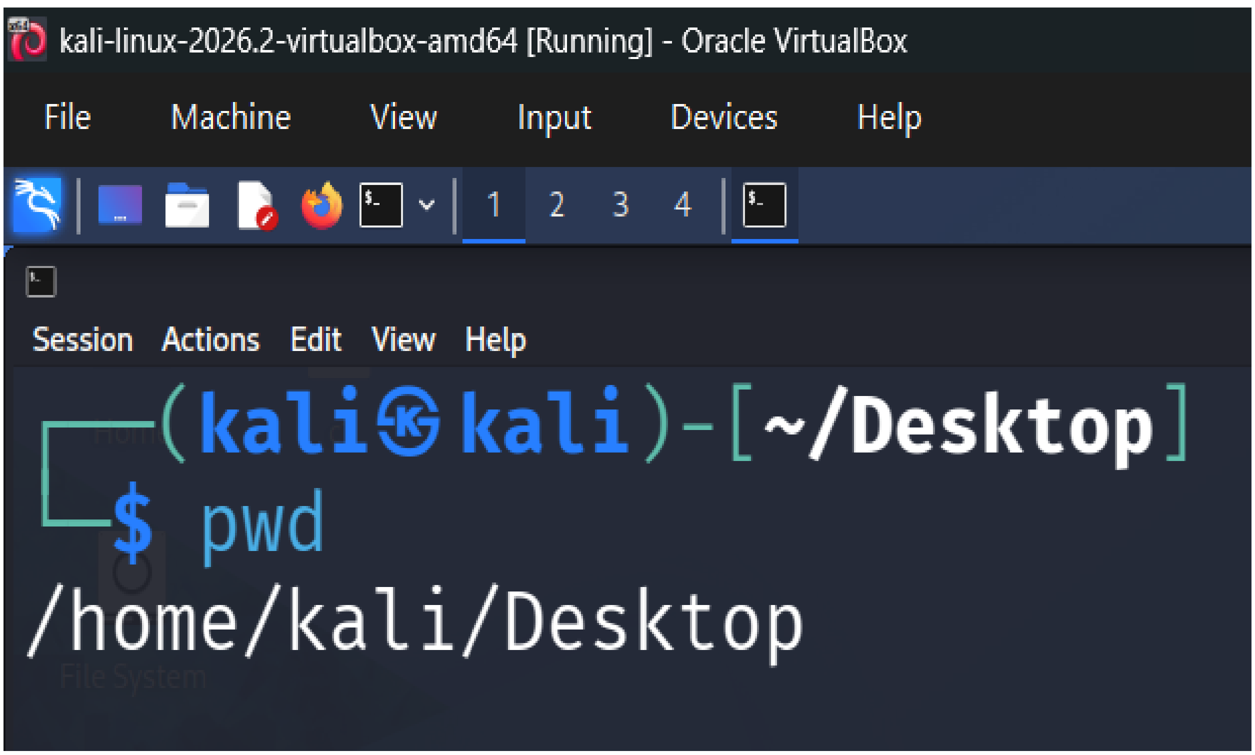
Now, the installation of VMM (Hypervisor Type-2 Oracle VirtualBox)& VM (Kali Linux) is completed.

Step 8 – Open terminal and create the following simple program, compile and execute to understand the working of the environment.

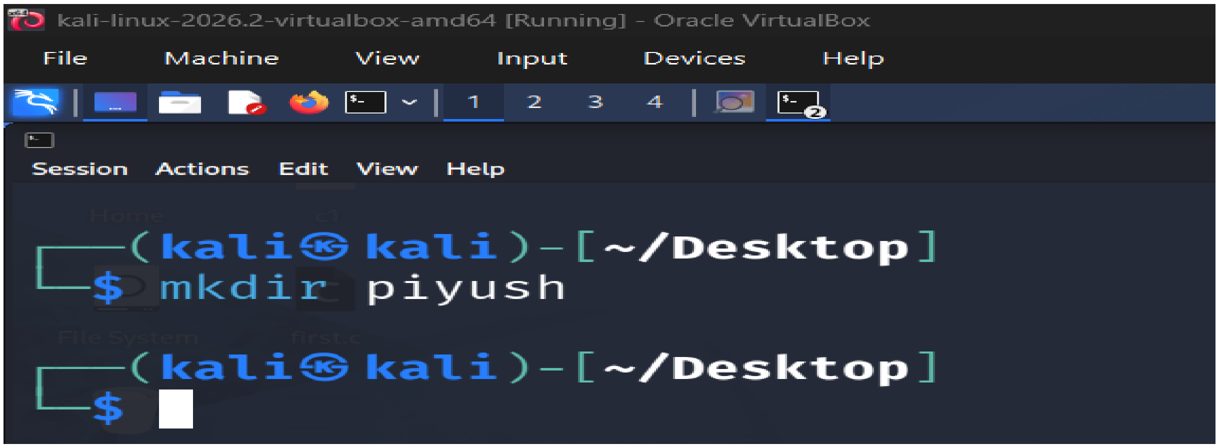
B) Working with Linux Commands

- **Terminal basics:**

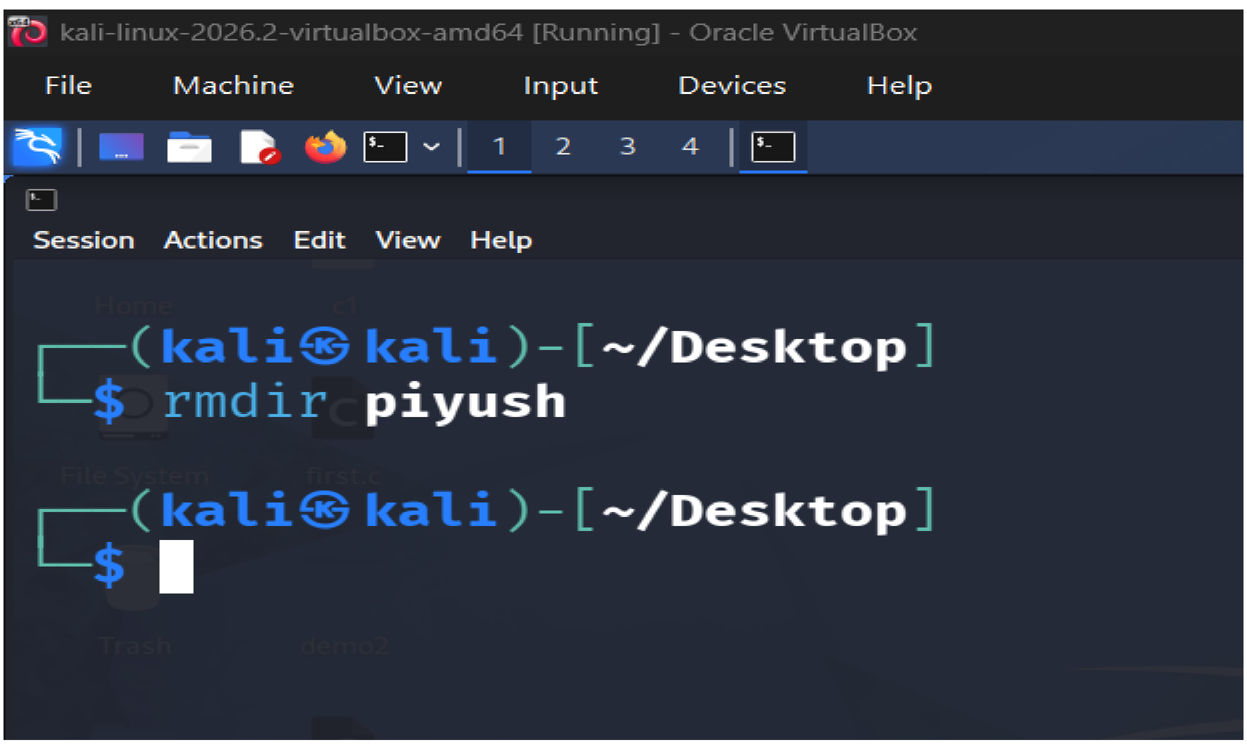
1) Pwd-Displays the absolute path of the current working directory.



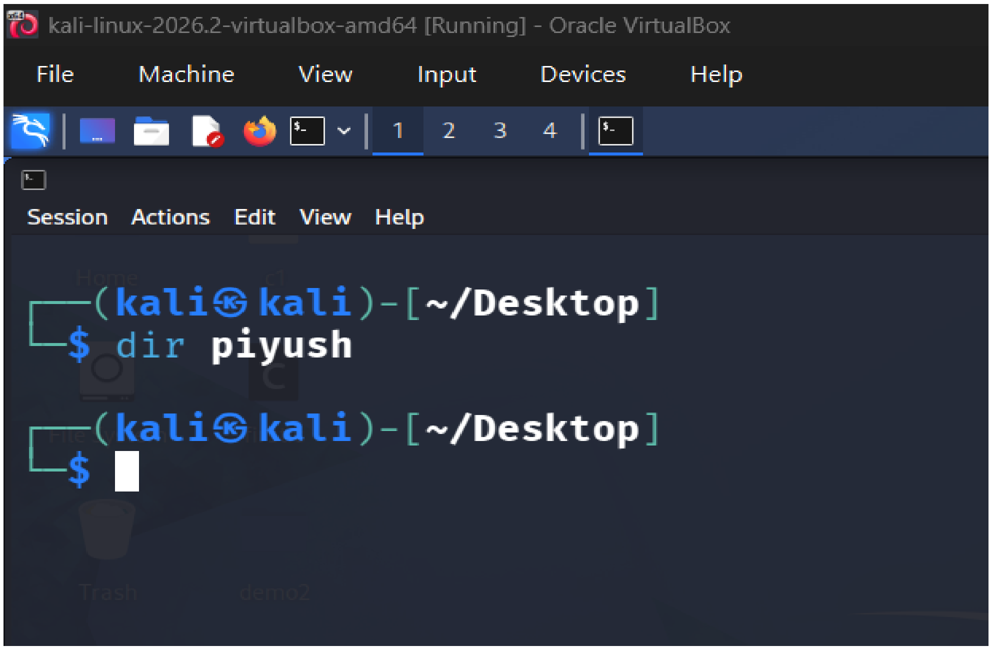
2) mkdir - Creates a new directory named Piyush.



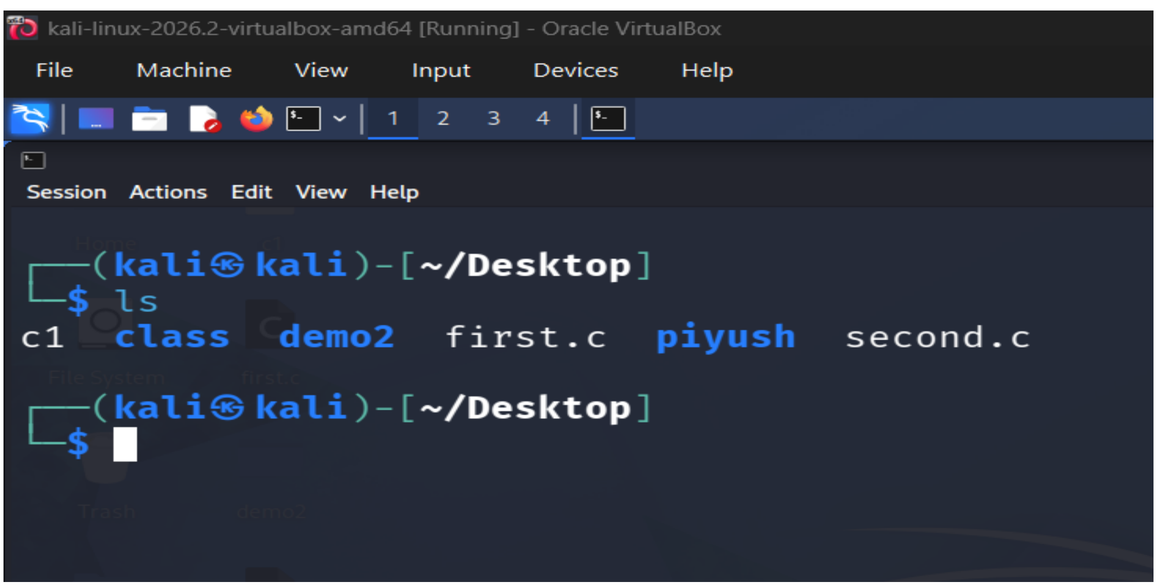
3) Rmdir-Removes an empty directory.



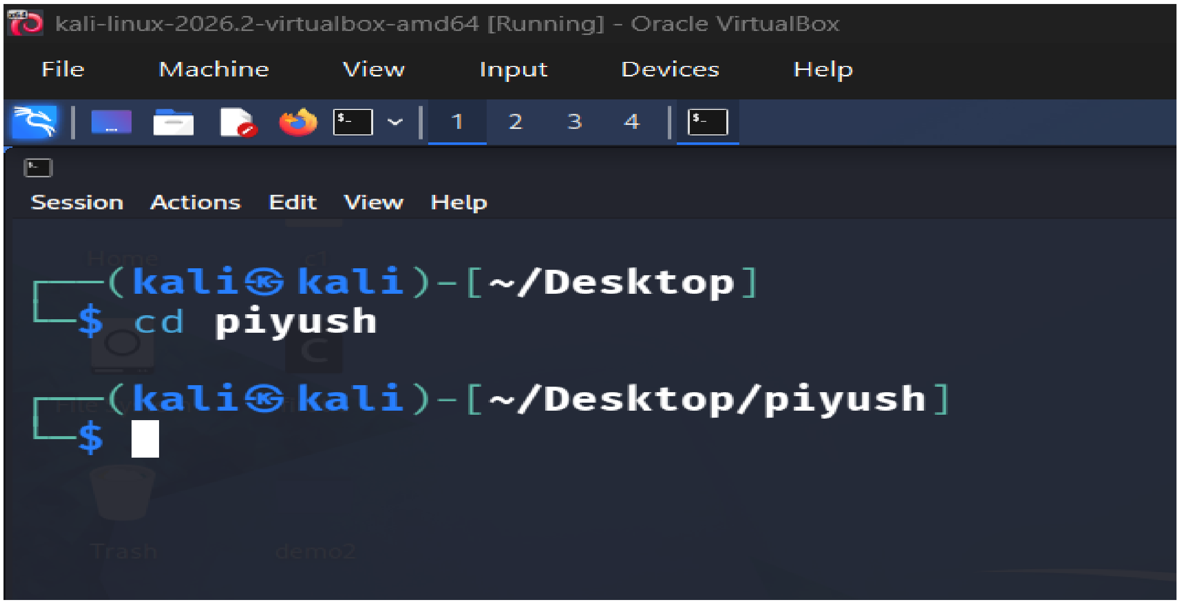
4) dir - Displays the files and directories in the current directory



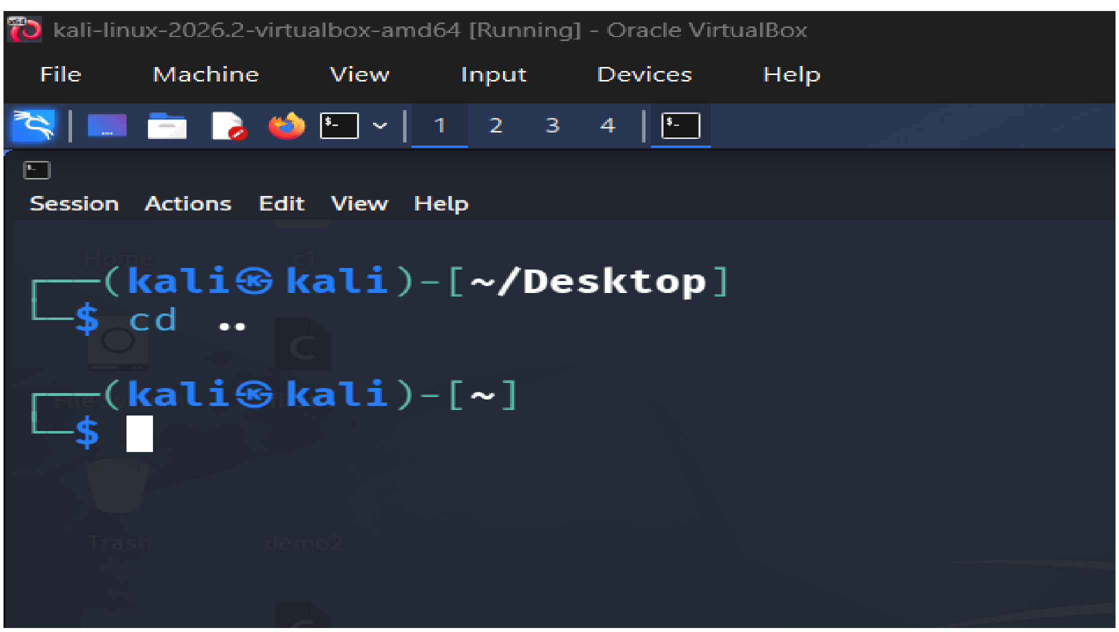
5) ls - Lists files and directories in the current directory.



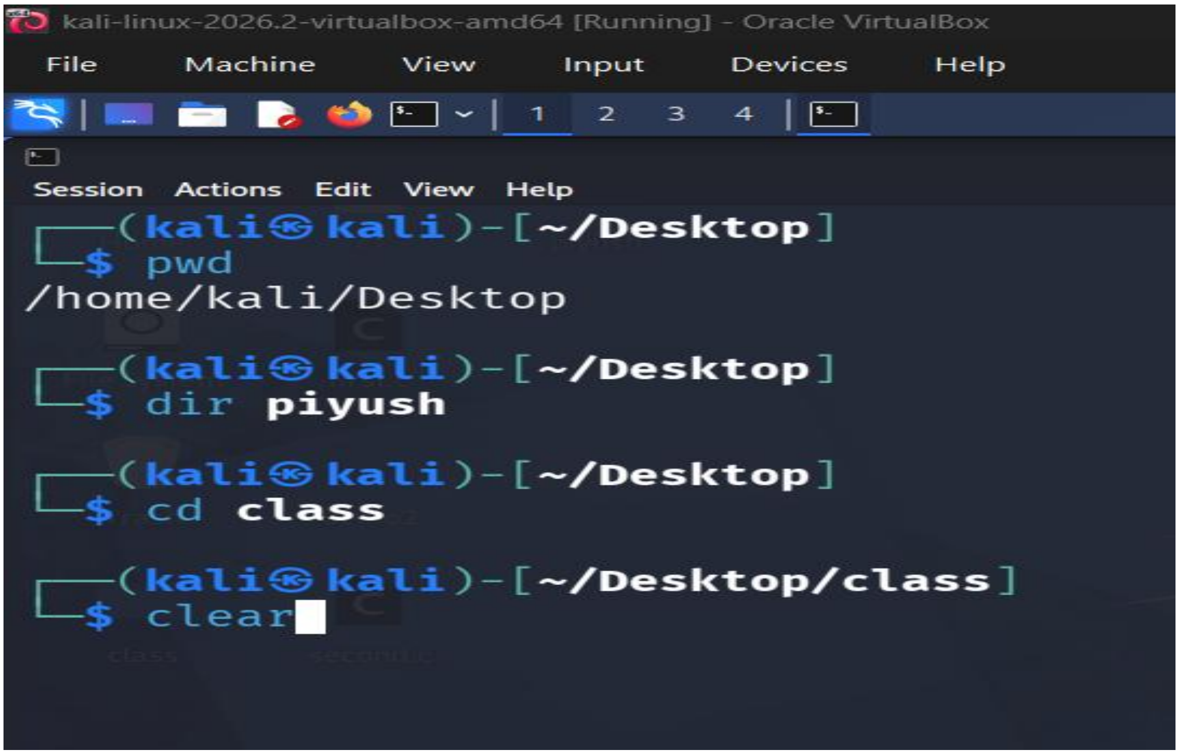
6) cd - Changes the current directory to piyush.



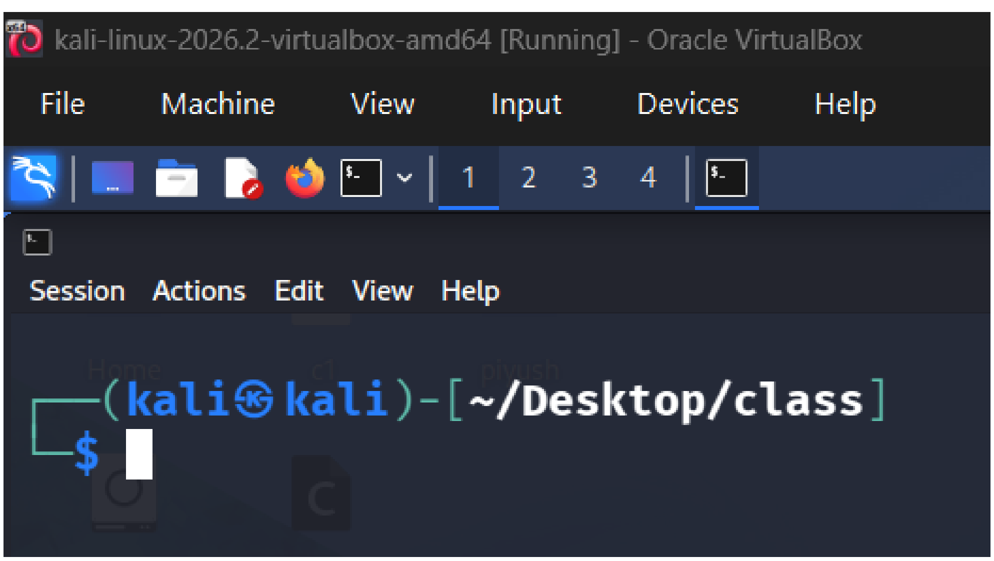
7) cd .. - Moves one level up to the parent directory.



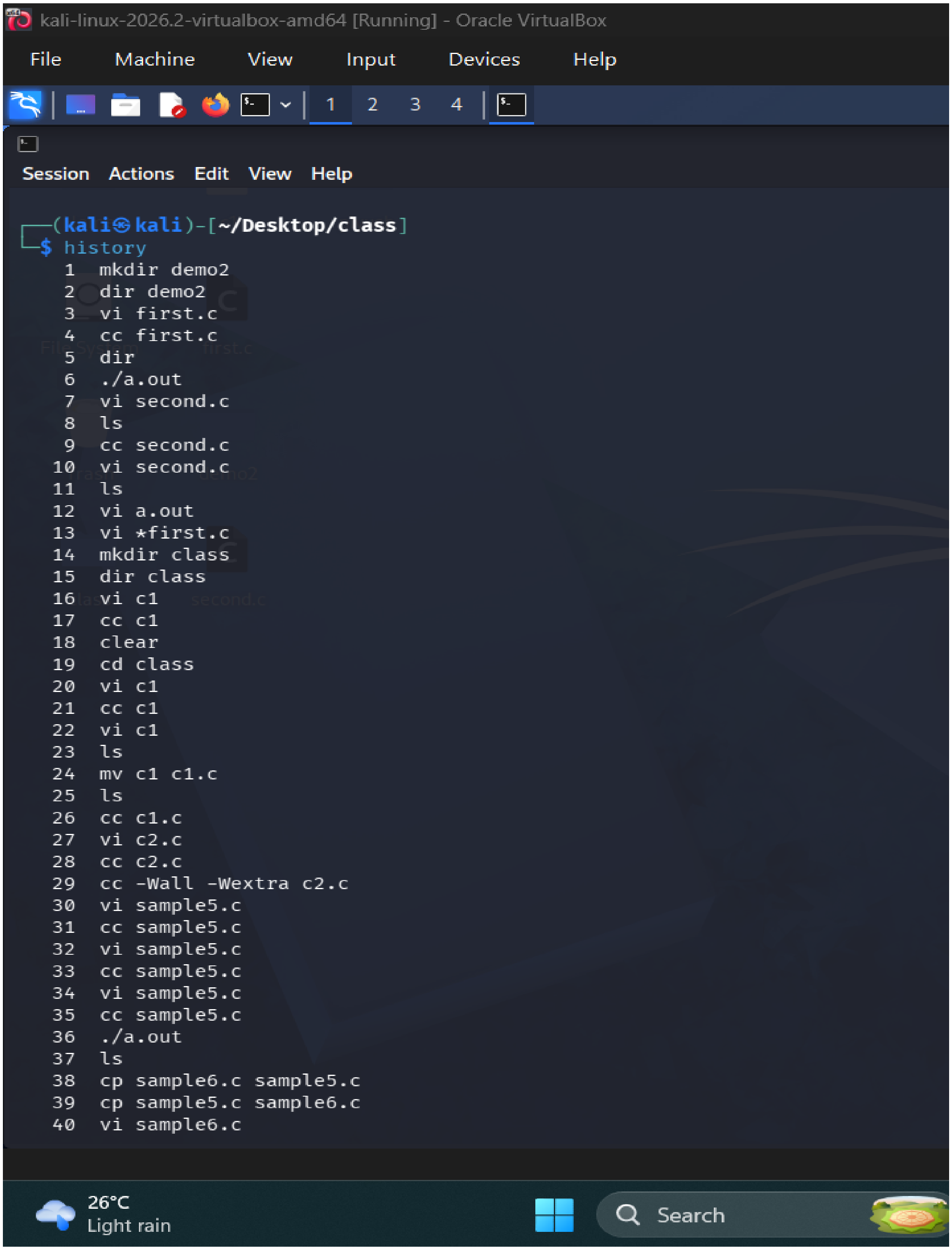
8) clear - Clears the terminal screen.

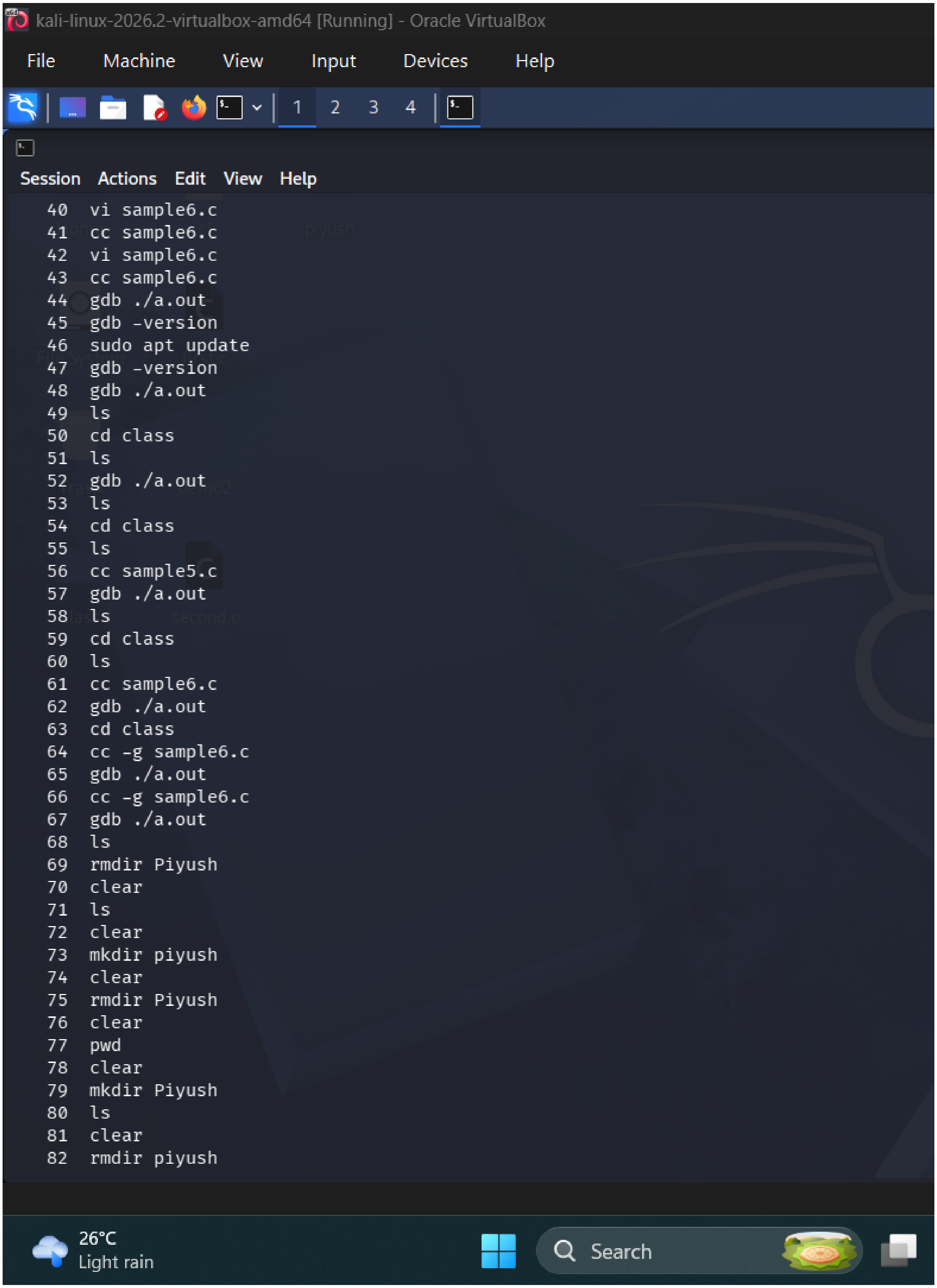


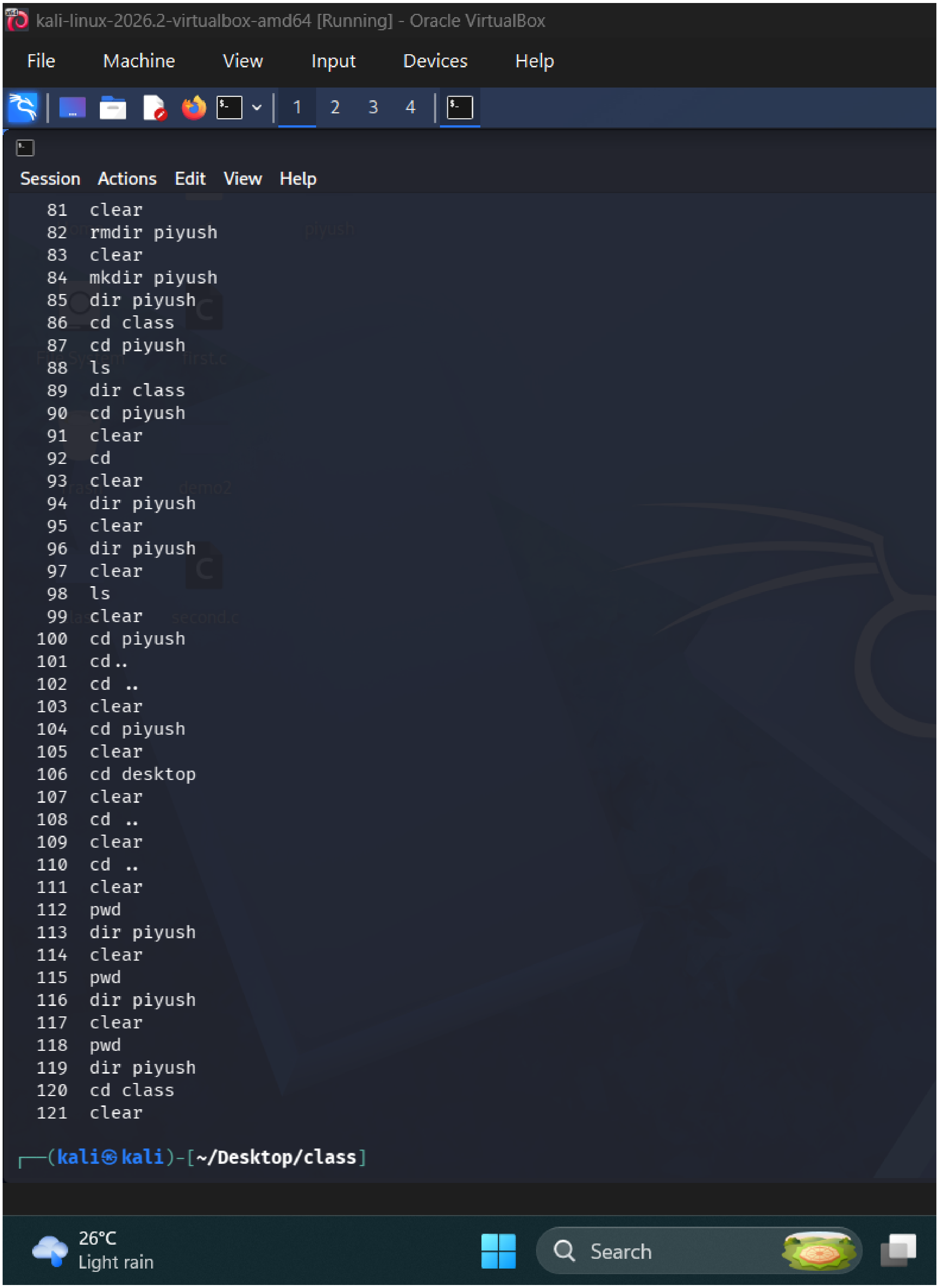
After executing the clear command.



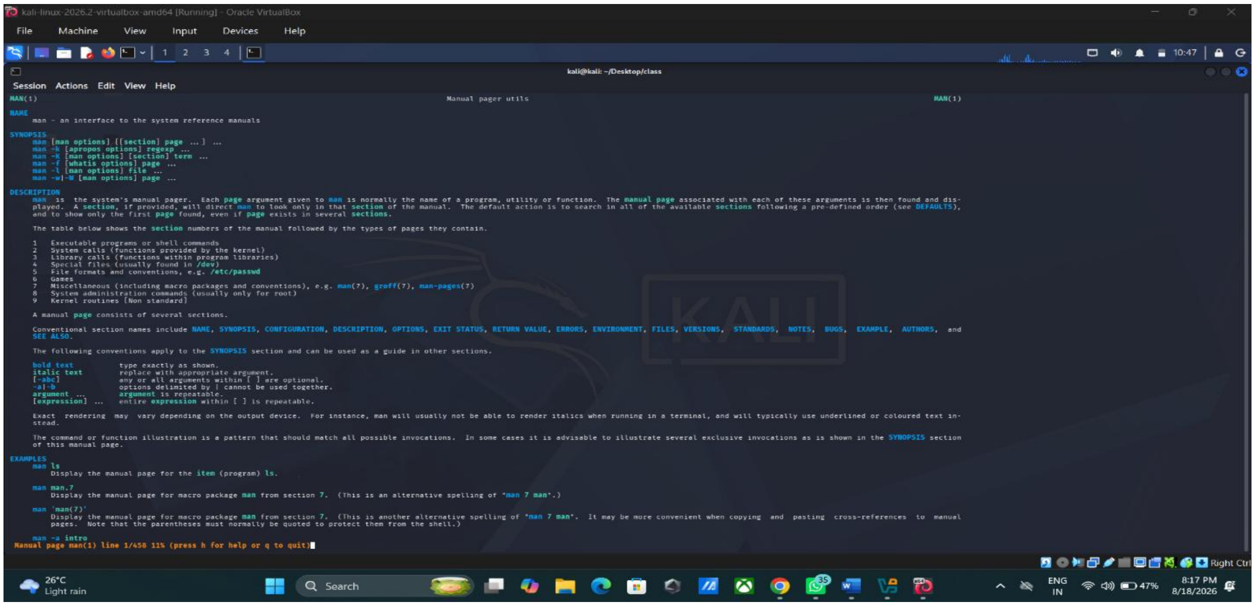
9) history – Displays all the previously executed commands.

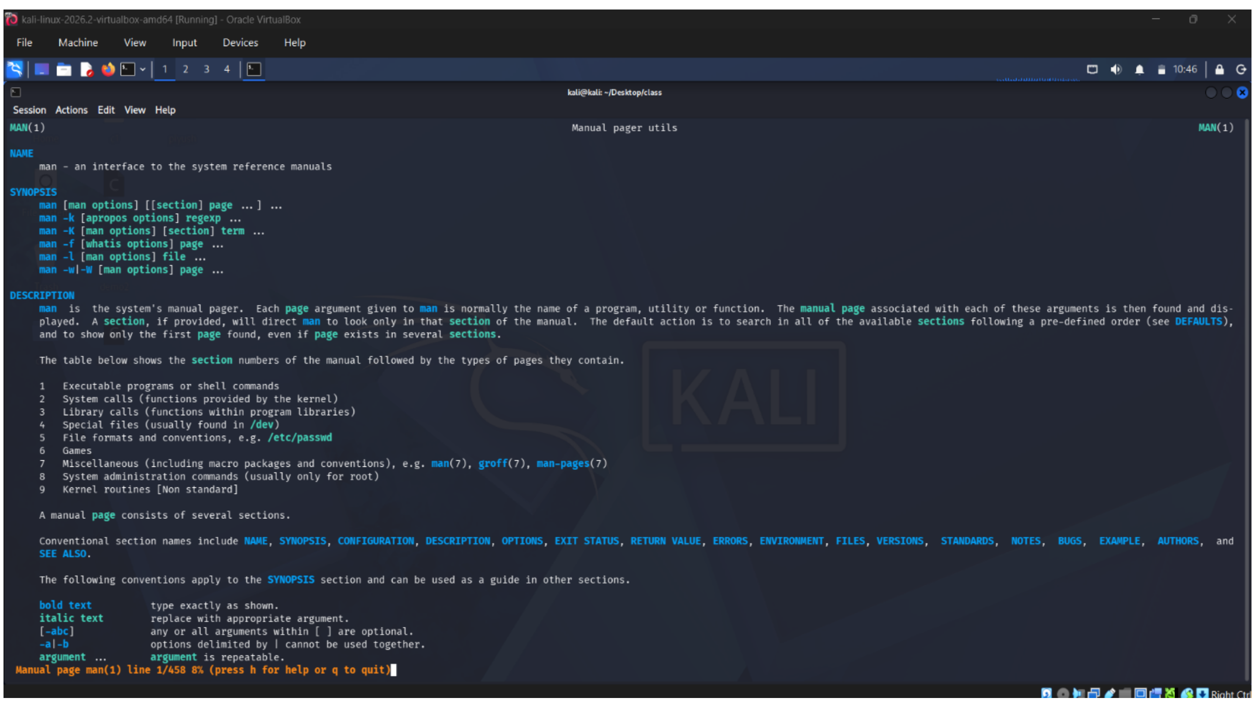


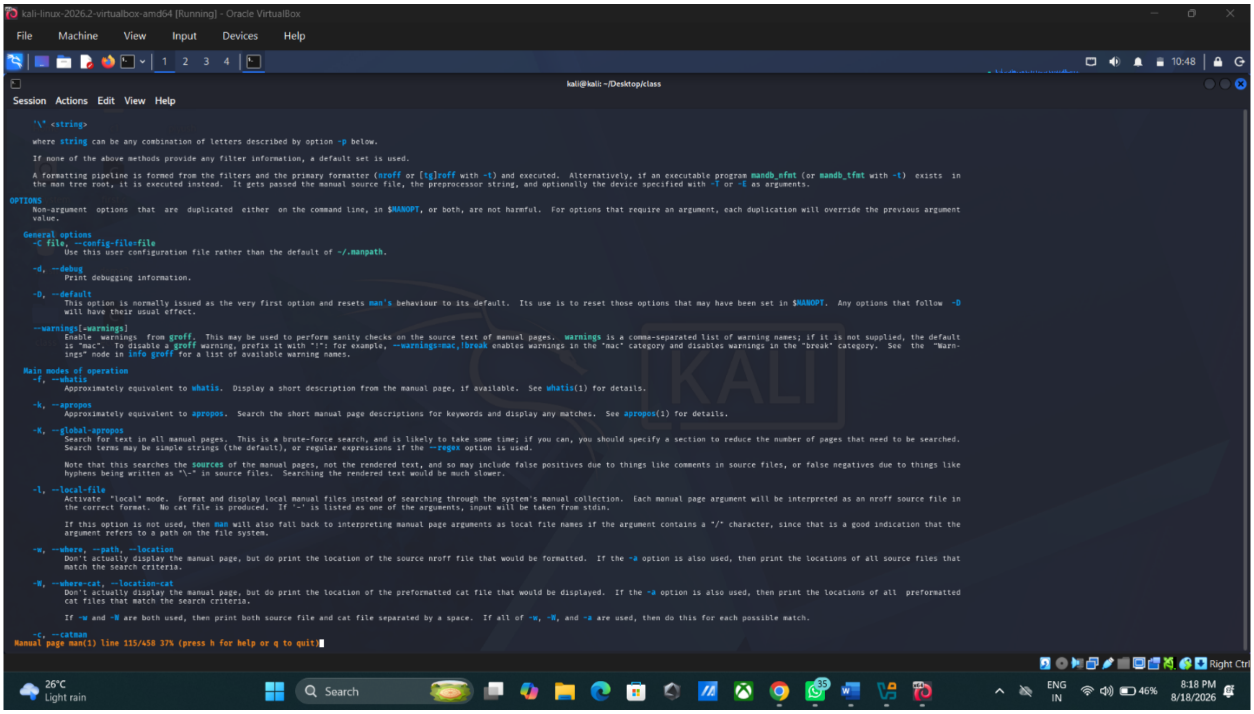

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
Session Actions Edit View Help
40 vi sample6.c
41 cc sample6.c
42 vi sample6.c
43 cc sample6.c
44 gdb ./a.out
45 gdb -version
46 sudo apt update
47 gdb -version
48 gdb ./a.out
49 ls
50 cd class
51 ls
52 gdb ./a.out
53 ls
54 cd class
55 ls
56 cc sample5.c
57 gdb ./a.out
58 ls
59 cd class
60 ls
61 cc sample6.c
62 gdb ./a.out
63 cd class
64 cc -g sample6.c
65 gdb ./a.out
66 cc -g sample6.c
67 gdb ./a.out
68 ls
69 rmdir Piyush
70 clear
71 ls
72 clear
73 mkdir piyush
74 clear
75 rmdir Piyush
76 clear
77 pwd
78 clear
79 mkdir Piyush
80 ls
81 clear
82 rmdir piyush
26°C
Light rain
Search


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
Session Actions Edit View Help
81 clear
82 rmdir piyush
83 clear
84 mkdir piyush
85 dir piyush
86 cd class
87 cd piyush
88 ls
89 dir class
90 cd piyush
91 clear
92 cd
93 clear
94 dir piyush
95 clear
96 dir piyush
97 clear
98 ls
99 clear
100 cd piyush
101 cd..
102 cd ..
103 clear
104 cd piyush
105 clear
106 cd desktop
107 clear
108 cd ..
109 clear
110 cd ..
111 clear
112 pwd
113 dir piyush
114 clear
115 pwd
116 dir piyush
117 clear
118 pwd
119 dir piyush
120 cd class
121 clear
┌──(kali㉿kali)-[~/Desktop/class]
26°C
Light rain
Search

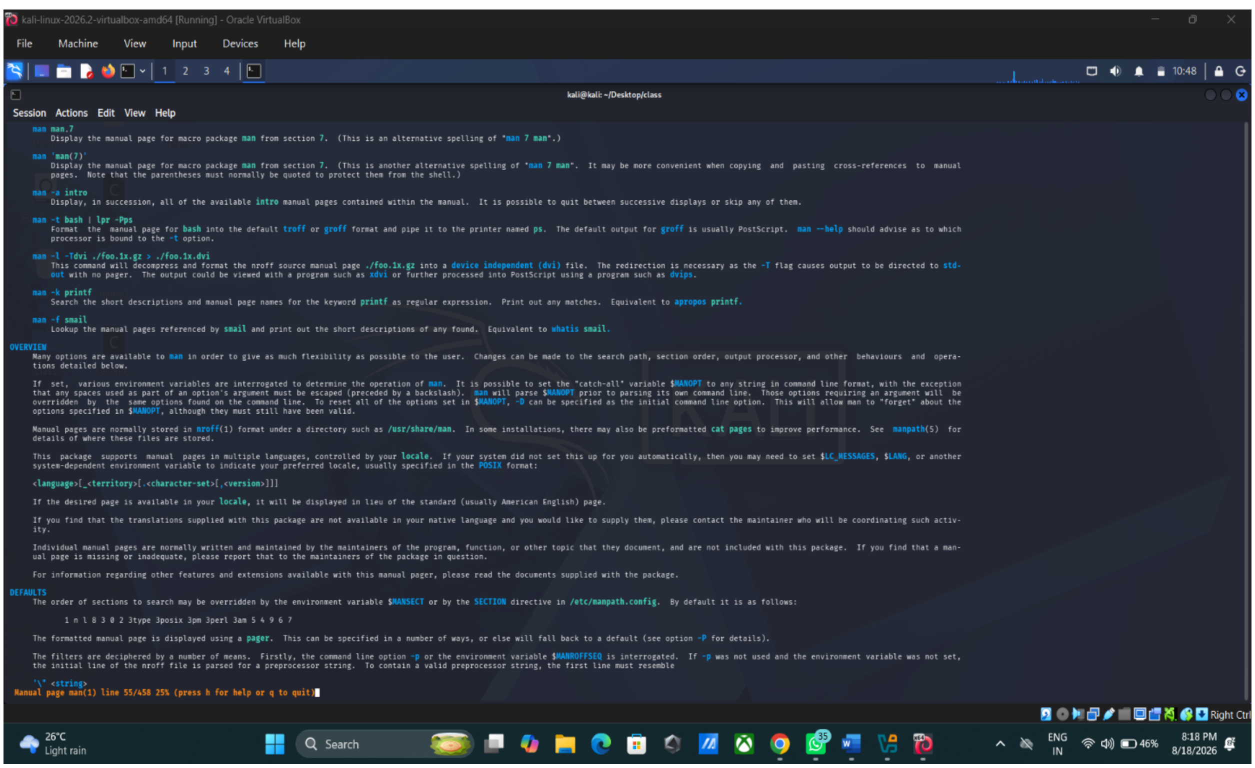
10) man - Displays the manual/help page for a command.



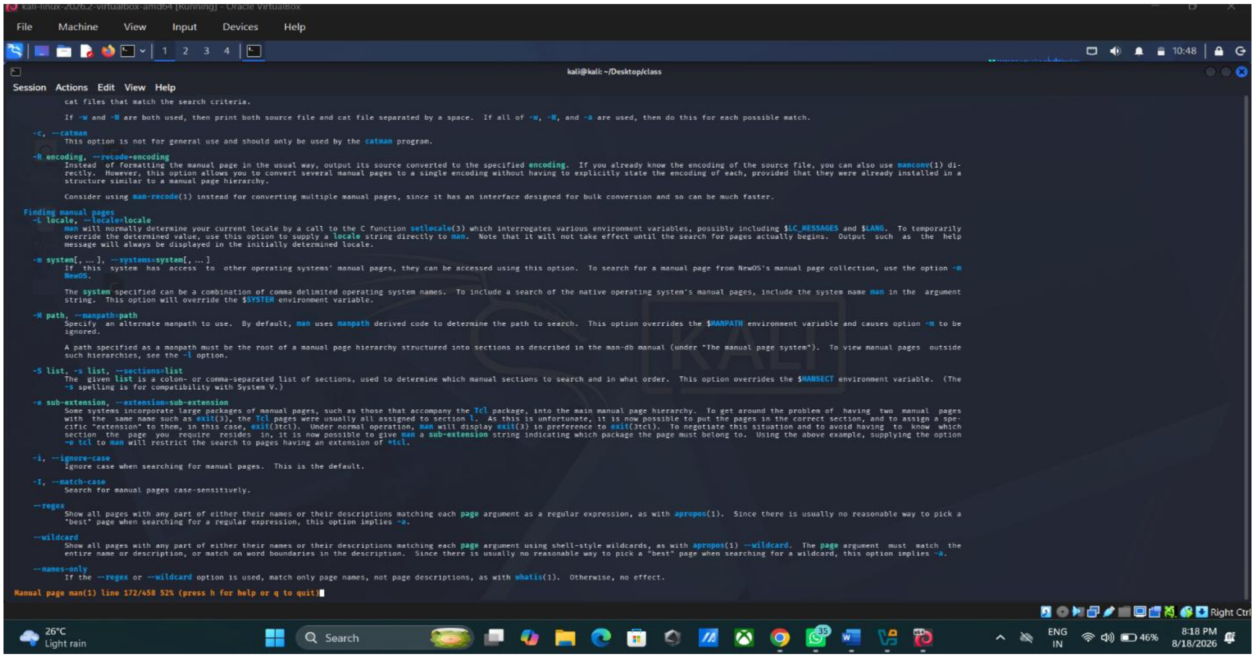


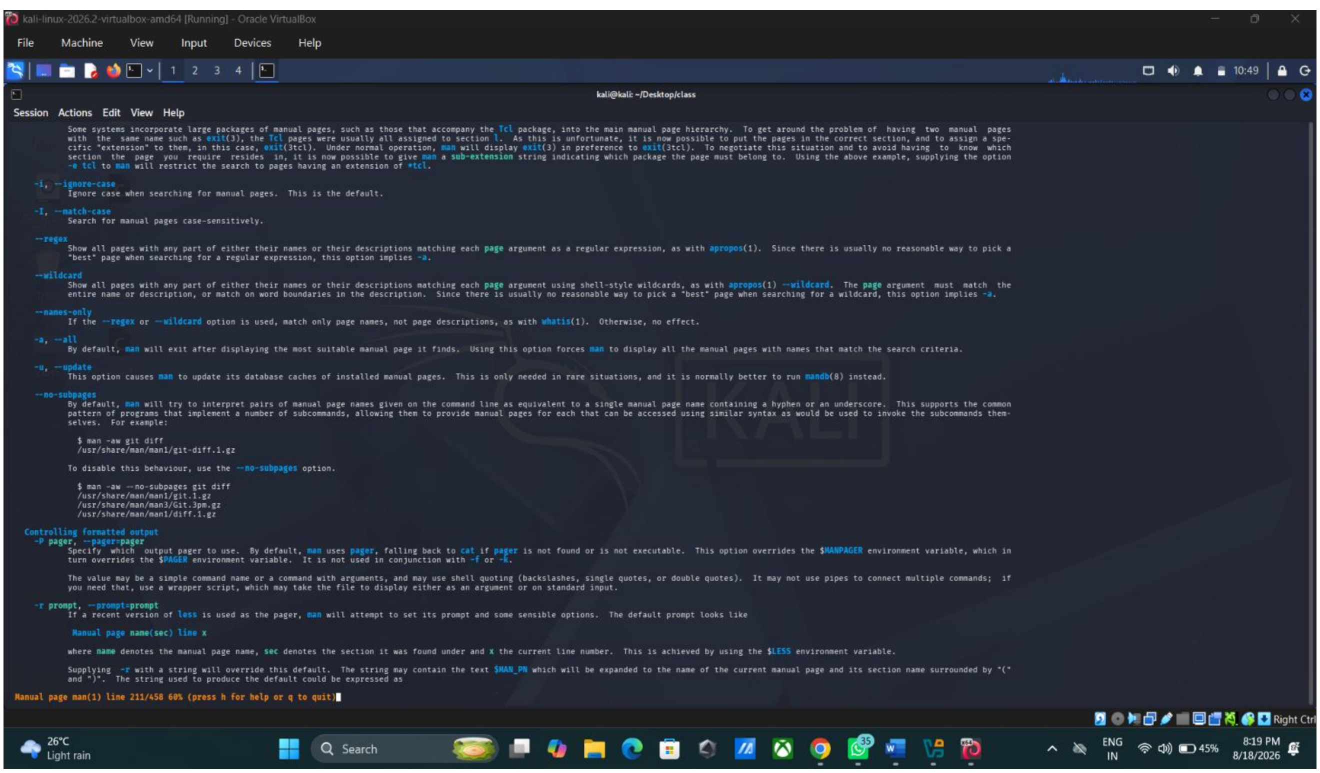



```
       '\" <string>

       where string can be any combination of letters described by option -p below.

       If none of the above methods provide any filter information, a default set is used.

       A formatting pipeline is formed from the filters and the primary formatter (nroff or [tg]roff with -t) and executed.  Alternatively, if an executable program mandb_nfmt (or mandb_tfmt with -t)  exists  in
       the man tree root, it is executed instead.  It gets passed the manual source file, the preprocessor string, and optionally the device specified with -T or -E as arguments.

OPTIONS
       Non-argument  options  that  are  duplicated  either  on the command line, in $MANOPT, or both, are not harmful.  For options that require an argument, each duplication will override the previous argument
       value.

   General options
       -C file, --config-file=file
              Use this user configuration file rather than the default of ~/.manpath.

       -d, --debug
              Print debugging information.

       -D, --default
              This option is normally issued as the very first option and resets man's behaviour to its default.  Its use is to reset those options that may have been set in $MANOPT.  Any options that follow  -D
              will have their usual effect.

       --warnings[=warnings]
              Enable  warnings  from groff.  This may be used to perform sanity checks on the source text of manual pages.  warnings is a comma-separated list of warning names; if it is not supplied, the default
              is "mac".  To disable a groff warning, prefix it with "!": for example, --warnings=mac,!break enables warnings in the "mac" category and disables warnings in the "break" category.  See  the  "Warn‐
              ings" node in info groff for a list of available warning names.

   Main modes of operation
       -f, --whatis
              Approximately equivalent to whatis.  Display a short description from the manual page, if available.  See whatis(1) for details.

       -k, --apropos
              Approximately equivalent to apropos.  Search the short manual page descriptions for keywords and display any matches.  See apropos(1) for details.

       -K, --global-apropos
              Search for text in all manual pages.  This is a brute-force search, and is likely to take some time; if you can, you should specify a section to reduce the number of pages that need to be searched.
              Search terms may be simple strings (the default), or regular expressions if the --regex option is used.

              Note that this searches the sources of the manual pages, not the rendered text, and so may include false positives due to things like comments in source files, or false negatives due to things like
              hyphens being written as "\-" in source files.  Searching the rendered text would be much slower.

       -l, --local-file
              Activate  "local" mode.  Format and display local manual files instead of searching through the system's manual collection.  Each manual page argument will be interpreted as an nroff source file in
              the correct format.  No cat file is produced.  If '-' is listed as one of the arguments, input will be taken from stdin.

              If this option is not used, then man will also fall back to interpreting manual page arguments as local file names if the argument contains a "/" character, since that is a good indication that the
              argument refers to a path on the file system.

       -w, --where, --path, --location
              Don't actually display the manual page, but do print the location of the source nroff file that would be formatted.  If the -a option is also used, then print the locations of all source files that
              match the search criteria.

       -W, --where-cat, --location-cat
              Don't actually display the manual page, but do print the location of the preformatted cat file that would be displayed.  If the -a option is also used, then print the locations of all  preformatted
              cat files that match the search criteria.

              If -w and -W are both used, then print both source file and cat file separated by a space.  If all of -w, -W, and -a are used, then do this for each possible match.

       -c, --catman
 Manual page man(1) line 115/458 37% (press h for help or q to quit)
```




```
       man man.7
           Display the manual page for macro package man from section 7.  (This is an alternative spelling of "man 7 man".)

       man 'man(7)'
           Display the manual page for macro package man from section 7.  (This is another alternative spelling of "man 7 man".  It may be more convenient when copying  and  pasting  cross-references  to  manual
           pages.  Note that the parentheses must normally be quoted to protect them from the shell.)

       man -a intro
           Display, in succession, all of the available intro manual pages contained within the manual.  It is possible to quit between successive displays or skip any of them.

       man -t bash | lpr -Pps
           Format  the  manual page for bash into the default troff or groff format and pipe it to the printer named ps.  The default output for groff is usually PostScript.  man --help should advise as to which
           processor is bound to the -t option.

       man -l -Tdvi ./foo.1x.gz > ./foo.1x.dvi
           This command will decompress and format the nroff source manual page ./foo.1x.gz into a device independent (dvi) file.  The redirection is necessary as the -T flag causes output to be directed to std‐
           out with no pager.  The output could be viewed with a program such as xdvi or further processed into PostScript using a program such as dvips.

       man -k printf
           Search the short descriptions and manual page names for the keyword printf as regular expression.  Print out any matches.  Equivalent to apropos printf.

       man -f smail
           Lookup the manual pages referenced by smail and print out the short descriptions of any found.  Equivalent to whatis smail.

OVERVIEW
       Many options are available to man in order to give as much flexibility as possible to the user.  Changes can be made to the search path, section order, output processor, and other  behaviours  and  opera‐
       tions detailed below.

       If  set,  various environment variables are interrogated to determine the operation of man.  It is possible to set the "catch-all" variable $MANOPT to any string in command line format, with the exception
       that any spaces used as part of an option's argument must be escaped (preceded by a backslash).  man will parse $MANOPT prior to parsing its own command line.  Those options requiring an argument will  be
       overridden  by  the  same options found on the command line.  To reset all of the options set in $MANOPT, -D can be specified as the initial command line option.  This will allow man to "forget" about the
       options specified in $MANOPT, although they must still have been valid.

       Manual pages are normally stored in nroff(1) format under a directory such as /usr/share/man.  In some installations, there may also be preformatted cat pages to improve performance.  See  manpath(5)  for
       details of where these files are stored.

       This  package  supports  manual  pages in multiple languages, controlled by your locale.  If your system did not set this up for you automatically, then you may need to set $LC_MESSAGES, $LANG, or another
       system-dependent environment variable to indicate your preferred locale, usually specified in the POSIX format:

       <language>[_<territory>[.<character-set>[,<version>]]]

       If the desired page is available in your locale, it will be displayed in lieu of the standard (usually American English) page.

       If you find that the translations supplied with this package are not available in your native language and you would like to supply them, please contact the maintainer who will be coordinating such activ‐
       ity.

       Individual manual pages are normally written and maintained by the maintainers of the program, function, or other topic that they document, and are not included with this package.  If you find that a man‐
       ual page is missing or inadequate, please report that to the maintainers of the package in question.

       For information regarding other features and extensions available with this manual pager, please read the documents supplied with the package.

DEFAULTS
       The order of sections to search may be overridden by the environment variable $MANSECT or by the SECTION directive in /etc/manpath.config.  By default it is as follows:

              1 n l 8 3 0 2 3type 3posix 3pm 3perl 3am 5 4 9 6 7

       The formatted manual page is displayed using a pager.  This can be specified in a number of ways, or else will fall back to a default (see option -P for details).

       The filters are deciphered by a number of means.  Firstly, the command line option -p or the environment variable $MANROFFSEQ is interrogated.  If -p was not used and the environment variable was not set,
       the initial line of the nroff file is parsed for a preprocessor string.  To contain a valid preprocessor string, the first line must resemble

       '\" <string>
 Manual page man(1) line 55/458 25% (press h for help or q to quit)
```


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~/Desktop/class
Session Actions Edit View Help
cat files that match the search criteria.
If -w and -W are both used, then print both source file and cat file separated by a space. If all of -w, -W, and -a are used, then do this for each possible match.
-c, --catman
This option is not for general use and should only be used by the catman program.
-R encoding, --recode=encoding
Instead of formatting the manual page in the usual way, output its source converted to the specified encoding. If you already know the encoding of the source file, you can also use manconv(1) directly. However, this option allows you to convert several manual pages to a single encoding without having to explicitly state the encoding of each, provided that they were already installed in a structure similar to a manual page hierarchy.
Consider using man-recode(1) instead for converting multiple manual pages, since it has an interface designed for bulk conversion and so can be much faster.
Finding manual pages
-L locale, --locale=locale
man will normally determine your current locale by a call to the C function setlocale(3) which interrogates various environment variables, possibly including $LC_MESSAGES and $LANG. To temporarily override the determined value, use this option to supply a locale string directly to man. Note that it will not take effect until the search for pages actually begins. Output such as the help message will always be displayed in the initially determined locale.
-m system[, ...], --systems=system[, ...]
If this system has access to other operating systems' manual pages, they can be accessed using this option. To search for a manual page from NewOS's manual page collection, use the option -m NewOS.
The system specified can be a combination of comma delimited operating system names. To include a search of the native operating system's manual pages, include the system name man in the argument string. This option will override the $SYSTEM environment variable.
-M path, --manpath=path
Specify an alternate manpath to use. By default, man uses manpath derived code to determine the path to search. This option overrides the $MANPATH environment variable and causes option -m to be ignored.
A path specified as a manpath must be the root of a manual page hierarchy structured into sections as described in the man-db manual (under "The manual page system"). To view manual pages outside such hierarchies, see the -l option.
-S list, -s list, --sections=list
The given list is a colon- or comma-separated list of sections, used to determine which manual sections to search and in what order. This option overrides the $MANSECT environment variable. (The -s spelling is for compatibility with System V.)
-e sub-extension, --extension=sub-extension
Some systems incorporate large packages of manual pages, such as those that accompany the Tcl package, into the main manual page hierarchy. To get around the problem of having two manual pages with the same name such as exit(3), the Tcl pages were usually all assigned to section l. As this is unfortunate, it is now possible to put the pages in the correct section, and to assign a specific "extension" to them, in this case, exit(3tcl). Under normal operation, man will display exit(3) in preference to exit(3tcl). To negotiate this situation and to avoid having to know which section the page you require resides in, it is now possible to give man a sub-extension string indicating which package the page must belong to. Using the above example, supplying the option -e tcl to man will restrict the search to pages having an extension of *tcl.
-i, --ignore-case
Ignore case when searching for manual pages. This is the default.
-I, --match-case
Search for manual pages case-sensitively.
--regex
Show all pages with any part of either their names or their descriptions matching each page argument as a regular expression, as with apropos(1). Since there is usually no reasonable way to pick a "best" page when searching for a regular expression, this option implies -a.
--wildcard
Show all pages with any part of either their names or their descriptions matching each page argument using shell-style wildcards, as with apropos(1) --wildcard. The page argument must match the entire name or description, or match on word boundaries in the description. Since there is usually no reasonable way to pick a "best" page when searching for a wildcard, this option implies -a.
--names-only
If the --regex or --wildcard option is used, match only page names, not page descriptions, as with whatis(1). Otherwise, no effect.
Manual page man(1) line 172/458 52% (press h for help or q to quit)
26°C
Light rain
Search
ENG
IN
46%
8:18 PM
8/18/2026


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~/Desktop/class
Session Actions Edit View Help
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Ignore case when searching for manual pages. This is the default.
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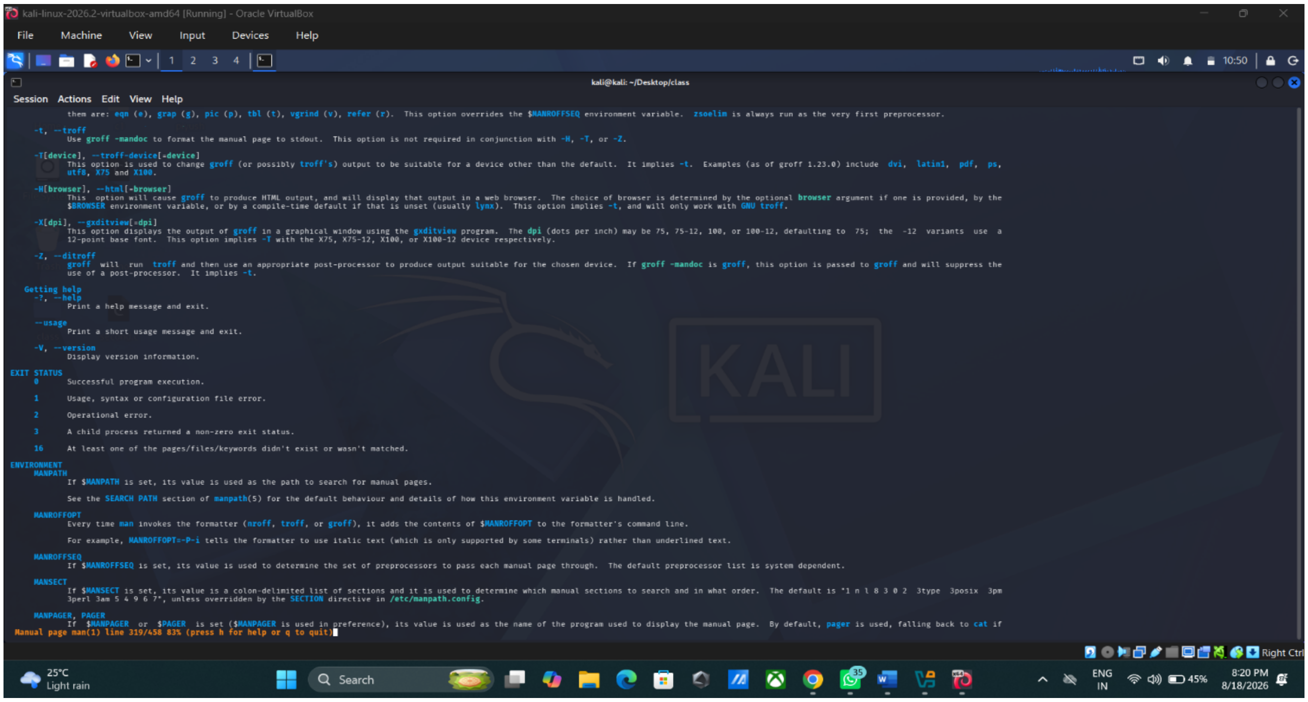
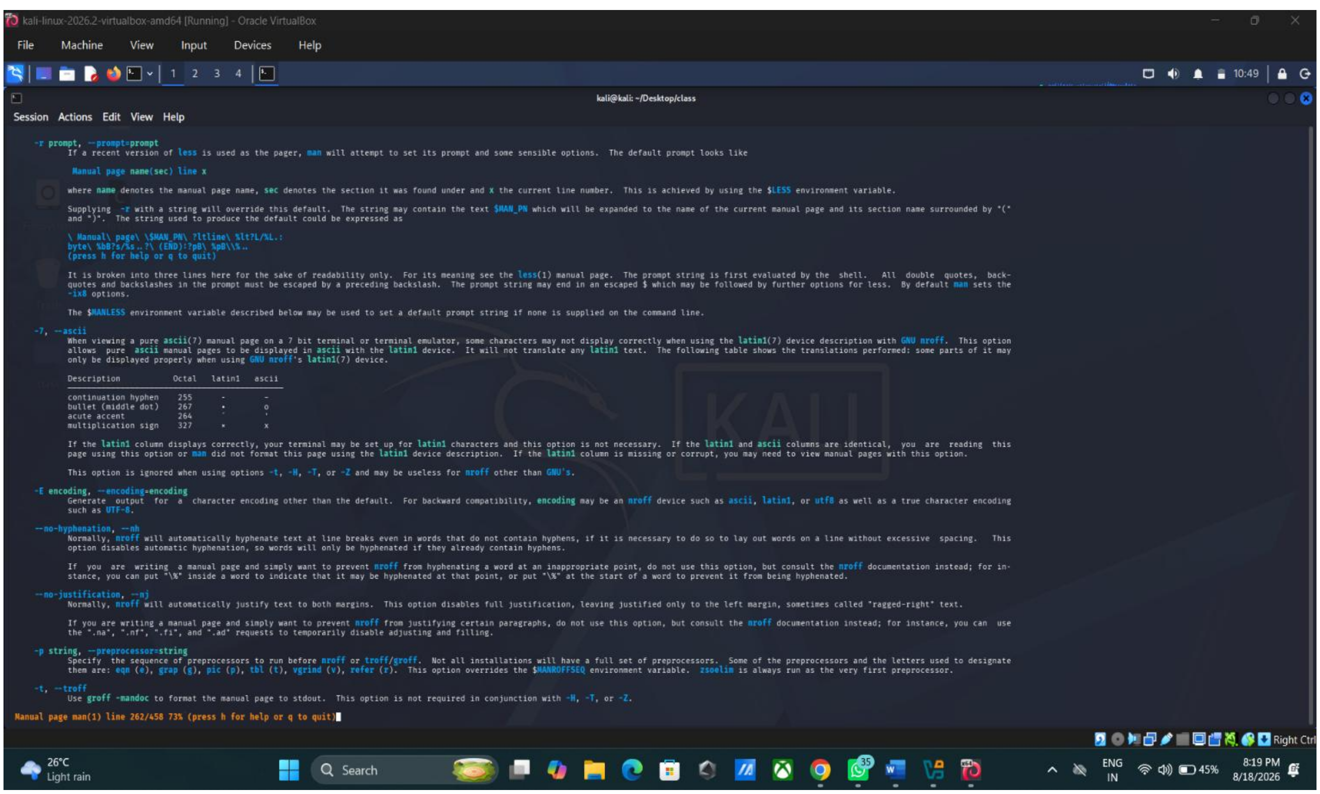
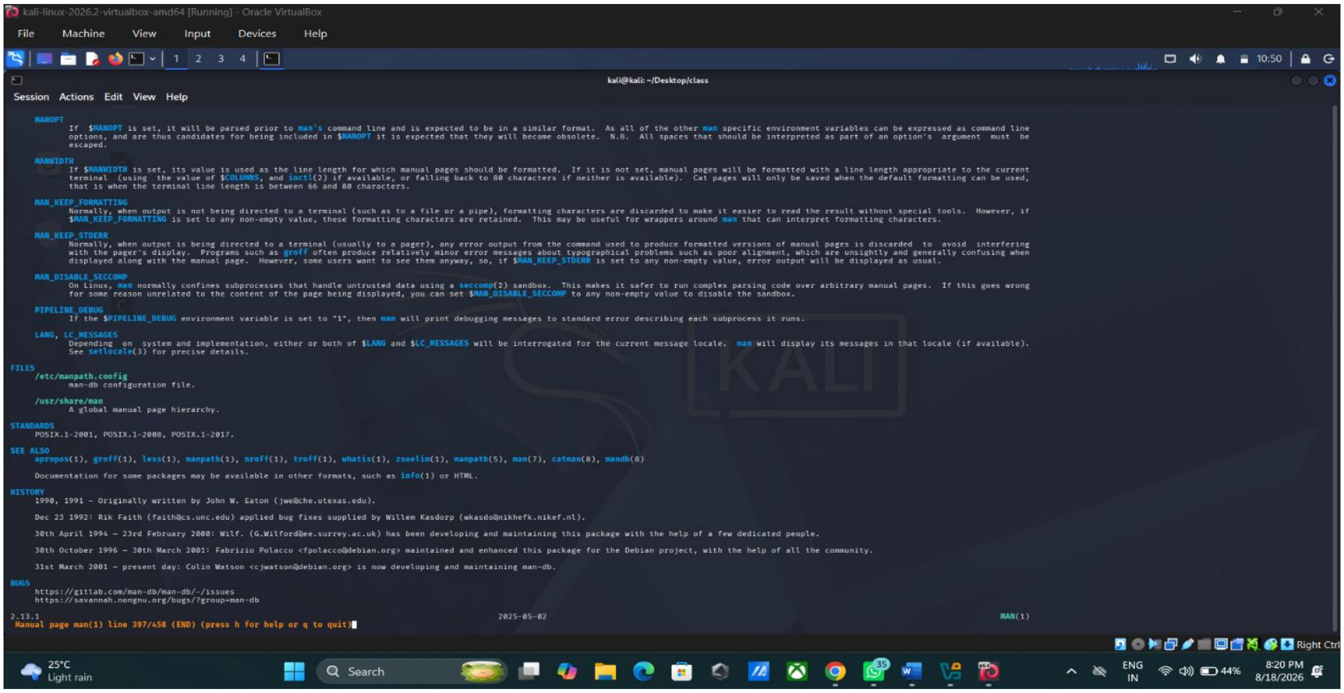
--wildcard
Show all pages with any part of either their names or their descriptions matching each page argument using shell-style wildcards, as with apropos(1) --wildcard. The page argument must match the entire name or description, or match on word boundaries in the description. Since there is usually no reasonable way to pick a "best" page when searching for a wildcard, this option implies -a.
--names-only
If the --regex or --wildcard option is used, match only page names, not page descriptions, as with whatis(1). Otherwise, no effect.
-a, --all
By default, man will exit after displaying the most suitable manual page it finds. Using this option forces man to display all the manual pages with names that match the search criteria.
-u, --update
This option causes man to update its database caches of installed manual pages. This is only needed in rare situations, and it is normally better to run mandb(8) instead.
--no-subpages
By default, man will try to interpret pairs of manual page names given on the command line as equivalent to a single manual page name containing a hyphen or an underscore. This supports the common pattern of programs that implement a number of subcommands, allowing them to provide manual pages for each that can be accessed using similar syntax as would be used to invoke the subcommands themselves. For example:
$ man -aw git diff
/usr/share/man/man1/git-diff.1.gz
To disable this behaviour, use the --no-subpages option.
$ man -aw --no-subpages git diff
/usr/share/man/man1/git.1.gz
/usr/share/man/man3/Git.3pm.gz
/usr/share/man/man1/diff.1.gz
Controlling formatted output
-P pager, --pager=pager
Specify which output pager to use. By default, man uses pager, falling back to cat if pager is not found or is not executable. This option overrides the $MANPAGER environment variable, which in turn overrides the $PAGER environment variable. It is not used in conjunction with -f or -k.
The value may be a simple command name or a command with arguments, and may use shell quoting (backslashes, single quotes, or double quotes). It may not use pipes to connect multiple commands; if you need that, use a wrapper script, which may take the file to display either as an argument or on standard input.
-r prompt, --prompt=prompt
If a recent version of less is used as the pager, man will attempt to set its prompt and some sensible options. The default prompt looks like
Manual page name(sec) line x
where name denotes the manual page name, sec denotes the section it was found under and x the current line number. This is achieved by using the $LESS environment variable.
Supplying -r with a string will override this default. The string may contain the text $MAN_PN which will be expanded to the name of the current manual page and its section name surrounded by "(" and ")". The string used to produce the default could be expressed as
Manual page man(1) line 211/458 60% (press h for help or q to quit)
26°C
Light rain
Search
ENG
IN
45%
8:19 PM
8/18/2026



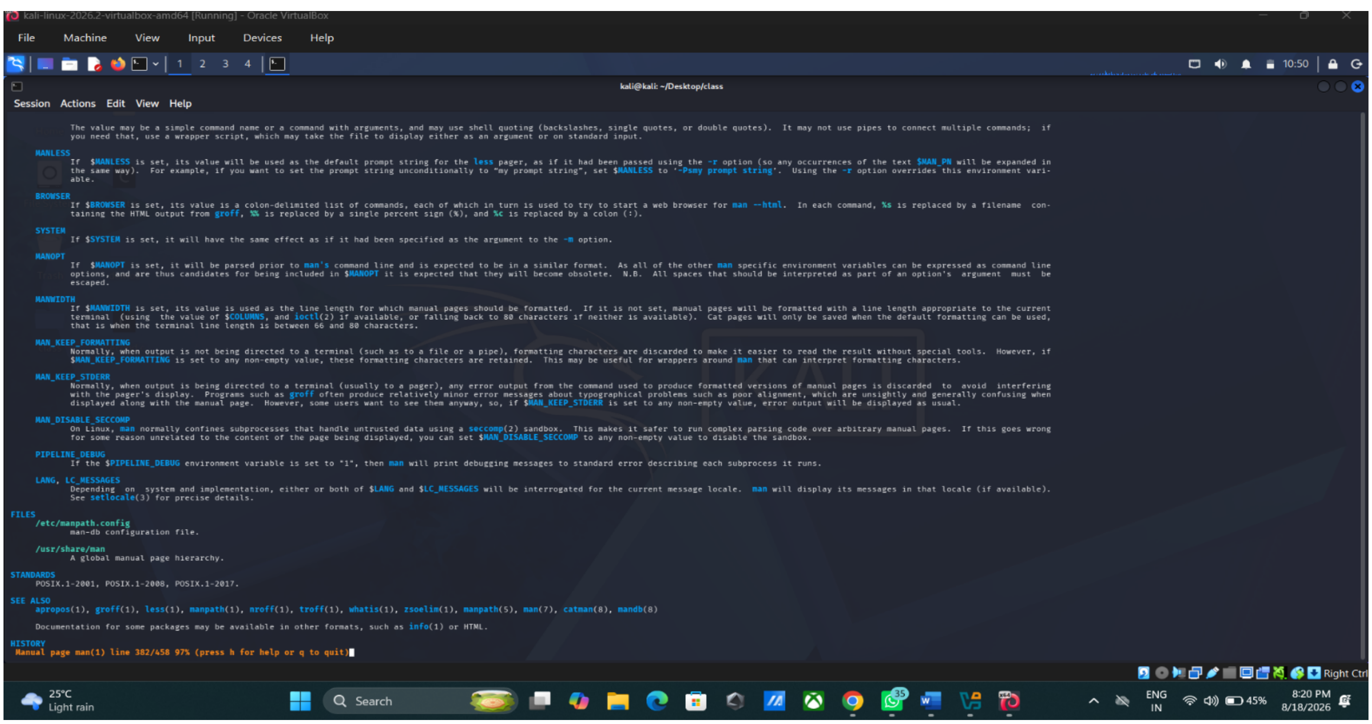

```
       them are: eqn (e), grap (g), pic (p), tbl (t), vgrind (v), refer (r).  This option overrides the $MANROFFSEQ environment variable.  zsoelim is always run as the very first preprocessor.

    -t, --troff
       Use groff -mandoc to format the manual page to stdout.  This option is not required in conjunction with -H, -T, or -Z.

    -T[device], --troff-device[=device]
       This option is used to change groff (or possibly troff's) output to be suitable for a device other than the default.  It implies -t.  Examples (as of groff 1.23.0) include  dvi,  latin1,  pdf,  ps,
       utf8, X75 and X100.

    -H[browser], --html[=browser]
       This  option will cause groff to produce HTML output, and will display that output in a web browser.  The choice of browser is determined by the optional browser argument if one is provided, by the
       $BROWSER environment variable, or by a compile-time default if that is unset (usually lynx).  This option implies -t, and will only work with GNU troff.

    -X[dpi], --gxditview[=dpi]
       This option displays the output of groff in a graphical window using the gxditview program.  The dpi (dots per inch) may be 75, 75-12, 100, or 100-12, defaulting to  75;  the  -12  variants  use  a
       12-point base font.  This option implies -T with the X75, X75-12, X100, or X100-12 device respectively.

    -Z, --ditroff
       groff  will  run  troff and then use an appropriate post-processor to produce output suitable for the chosen device.  If groff -mandoc is groff, this option is passed to groff and will suppress the
       use of a post-processor.  It implies -t.

  Getting help
    -?, --help
       Print a help message and exit.

    --usage
       Print a short usage message and exit.

    -V, --version
       Display version information.

EXIT STATUS
    0      Successful program execution.

    1      Usage, syntax or configuration file error.

    2      Operational error.

    3      A child process returned a non-zero exit status.

    16     At least one of the pages/files/keywords didn't exist or wasn't matched.

ENVIRONMENT
    MANPATH
       If $MANPATH is set, its value is used as the path to search for manual pages.

       See the SEARCH PATH section of manpath(5) for the default behaviour and details of how this environment variable is handled.

    MANROFFOPT
       Every time man invokes the formatter (nroff, troff, or groff), it adds the contents of $MANROFFOPT to the formatter's command line.

       For example, MANROFFOPT=-P-i tells the formatter to use italic text (which is only supported by some terminals) rather than underlined text.

    MANROFFSEQ
       If $MANROFFSEQ is set, its value is used to determine the set of preprocessors to pass each manual page through.  The default preprocessor list is system dependent.

    MANSECT
       If $MANSECT is set, its value is a colon-delimited list of sections and it is used to determine which manual sections to search and in what order.  The default is "1 n l 8 3 0 2  3type  3posix  3pm
       3perl 3am 5 4 9 6 7", unless overridden by the SECTION directive in /etc/manpath.config.

    MANPAGER, PAGER
       If  $MANPAGER  or  $PAGER  is set ($MANPAGER is used in preference), its value is used as the name of the program used to display the manual page.  By default, pager is used, falling back to cat if
 Manual page man(1) line 319/458 83% (press h for help or q to quit)
```




```
    -r prompt, --prompt=prompt
       If a recent version of less is used as the pager, man will attempt to set its prompt and some sensible options.  The default prompt looks like

        Manual page name(sec) line x

       where name denotes the manual page name, sec denotes the section it was found under and x the current line number.  This is achieved by using the $LESS environment variable.

       Supplying  -r with a string will override this default.  The string may contain the text $MAN_PN which will be expanded to the name of the current manual page and its section name surrounded by "("
       and ")".  The string used to produce the default could be expressed as

       \ Manual\ page\ \$MAN_PN\ ?ltline\ %lt?L/%L.:
       byte\ %bB?s/%s..?\ (END):?pB\ %pB\\%..
       (press h for help or q to quit)

       It is broken into three lines here for the sake of readability only.  For its meaning see the less(1) manual page.  The prompt string is first evaluated by the  shell.   All  double  quotes,  back-
       quotes and backslashes in the prompt must be escaped by a preceding backslash.  The prompt string may end in an escaped $ which may be followed by further options for less.  By default man sets the
       -ix8 options.

       The $MANLESS environment variable described below may be used to set a default prompt string if none is supplied on the command line.

    -7, --ascii
       When viewing a pure ascii(7) manual page on a 7 bit terminal or terminal emulator, some characters may not display correctly when using the latin1(7) device description with GNU nroff.  This option
       allows  pure  ascii manual pages to be displayed in ascii with the latin1 device.  It will not translate any latin1 text.  The following table shows the translations performed: some parts of it may
       only be displayed properly when using GNU nroff's latin1(7) device.

       Description            Octal   latin1   ascii
       ─────────────────────────────────────────────
       continuation hyphen    255       -        -
       bullet (middle dot)    267       ·        o
       acute accent           264       ´        '
       multiplication sign    327       ×        x

       If the latin1 column displays correctly, your terminal may be set up for latin1 characters and this option is not necessary.  If the latin1 and ascii columns are identical,  you  are  reading  this
       page using this option or man did not format this page using the latin1 device description.  If the latin1 column is missing or corrupt, you may need to view manual pages with this option.

       This option is ignored when using options -t, -H, -T, or -Z and may be useless for nroff other than GNU's.

    -E encoding, --encoding=encoding
       Generate  output  for  a  character encoding other than the default.  For backward compatibility, encoding may be an nroff device such as ascii, latin1, or utf8 as well as a true character encoding
       such as UTF-8.

    --no-hyphenation, --nh
       Normally, nroff will automatically hyphenate text at line breaks even in words that do not contain hyphens, if it is necessary to do so to lay out words on a line without excessive  spacing.   This
       option disables automatic hyphenation, so words will only be hyphenated if they already contain hyphens.

       If  you  are  writing  a manual page and simply want to prevent nroff from hyphenating a word at an inappropriate point, do not use this option, but consult the nroff documentation instead; for in-
       stance, you can put "\%" inside a word to indicate that it may be hyphenated at that point, or put "\%" at the start of a word to prevent it from being hyphenated.

    --no-justification, --nj
       Normally, nroff will automatically justify text to both margins.  This option disables full justification, leaving justified only to the left margin, sometimes called "ragged-right" text.

       If you are writing a manual page and simply want to prevent nroff from justifying certain paragraphs, do not use this option, but consult the nroff documentation instead; for instance, you can  use
       the ".na", ".nf", ".fi", and ".ad" requests to temporarily disable adjusting and filling.

    -p string, --preprocessor=string
       Specify  the sequence of preprocessors to run before nroff or troff/groff.  Not all installations will have a full set of preprocessors.  Some of the preprocessors and the letters used to designate
       them are: eqn (e), grap (g), pic (p), tbl (t), vgrind (v), refer (r).  This option overrides the $MANROFFSEQ environment variable.  zsoelim is always run as the very first preprocessor.

    -t, --troff
       Use groff -mandoc to format the manual page to stdout.  This option is not required in conjunction with -H, -T, or -Z.

 Manual page man(1) line 262/458 73% (press h for help or q to quit)
```



```
       MANOPT
              If  $MANOPT is set, it will be parsed prior to man's command line and is expected to be in a similar format.  As all of the other man specific environment variables can be expressed as command line
              options, and are thus candidates for being included in $MANOPT it is expected that they will become obsolete.  N.B.  All spaces that should be interpreted as part of an option's  argument  must  be
              escaped.

       MANWIDTH
              If $MANWIDTH is set, its value is used as the line length for which manual pages should be formatted.  If it is not set, manual pages will be formatted with a line length appropriate to the current
              terminal  (using  the value of $COLUMNS, and ioctl(2) if available, or falling back to 80 characters if neither is available).  Cat pages will only be saved when the default formatting can be used,
              that is when the terminal line length is between 66 and 80 characters.

       MAN_KEEP_FORMATTING
              Normally, when output is not being directed to a terminal (such as to a file or a pipe), formatting characters are discarded to make it easier to read the result without special tools.  However, if
              $MAN_KEEP_FORMATTING is set to any non-empty value, these formatting characters are retained.  This may be useful for wrappers around man that can interpret formatting characters.

       MAN_KEEP_STDERR
              Normally, when output is being directed to a terminal (usually to a pager), any error output from the command used to produce formatted versions of manual pages is discarded  to  avoid  interfering
              with the pager's display.  Programs such as groff often produce relatively minor error messages about typographical problems such as poor alignment, which are unsightly and generally confusing when
              displayed along with the manual page.  However, some users want to see them anyway, so, if $MAN_KEEP_STDERR is set to any non-empty value, error output will be displayed as usual.

       MAN_DISABLE_SECCOMP
              On Linux, man normally confines subprocesses that handle untrusted data using a seccomp(2) sandbox.  This makes it safer to run complex parsing code over arbitrary manual pages.  If this goes wrong
              for some reason unrelated to the content of the page being displayed, you can set $MAN_DISABLE_SECCOMP to any non-empty value to disable the sandbox.

       PIPELINE_DEBUG
              If the $PIPELINE_DEBUG environment variable is set to "1", then man will print debugging messages to standard error describing each subprocess it runs.

       LANG, LC_MESSAGES
              Depending  on  system and implementation, either or both of $LANG and $LC_MESSAGES will be interrogated for the current message locale.  man will display its messages in that locale (if available).
              See setlocale(3) for precise details.

FILES
       /etc/manpath.config
              man-db configuration file.

       /usr/share/man
              A global manual page hierarchy.

STANDARDS
       POSIX.1-2001, POSIX.1-2008, POSIX.1-2017.

SEE ALSO
       apropos(1), groff(1), less(1), manpath(1), nroff(1), troff(1), whatis(1), zsoelim(1), manpath(5), man(7), catman(8), mandb(8)

       Documentation for some packages may be available in other formats, such as info(1) or HTML.

HISTORY
       1990, 1991 – Originally written by John W. Eaton (jwe@che.utexas.edu).

       Dec 23 1992: Rik Faith (faith@cs.unc.edu) applied bug fixes supplied by Willem Kasdorp (wkasdo@nikhefk.nikef.nl).

       30th April 1994 – 23rd February 2000: Wilf. (G.Wilford@ee.surrey.ac.uk) has been developing and maintaining this package with the help of a few dedicated people.

       30th October 1996 – 30th March 2001: Fabrizio Polacco <fpolacco@debian.org> maintained and enhanced this package for the Debian project, with the help of all the community.

       31st March 2001 – present day: Colin Watson <cjwatson@debian.org> is now developing and maintaining man-db.

BUGS
       https://gitlab.com/man-db/man-db/-/issues
       https://savannah.nongnu.org/bugs/?group=man-db

2.13.1                                        2025-05-02                                        MAN(1)
 Manual page man(1) line 397/458 (END) (press h for help or q to quit)
```



```
              The value may be a simple command name or a command with arguments, and may use shell quoting (backslashes, single quotes, or double quotes).  It may not use pipes to connect multiple commands;  if
              you need that, use a wrapper script, which may take the file to display either as an argument or on standard input.

       MANLESS
              If  $MANLESS is set, its value will be used as the default prompt string for the less pager, as if it had been passed using the -r option (so any occurrences of the text $MAN_PN will be expanded in
              the same way).  For example, if you want to set the prompt string unconditionally to "my prompt string", set $MANLESS to '-Psmy prompt string'.  Using the -r option overrides this environment vari-
              able.

       BROWSER
              If $BROWSER is set, its value is a colon-delimited list of commands, each of which in turn is used to try to start a web browser for man --html.  In each command, %s is replaced by a filename  con-
              taining the HTML output from groff, %% is replaced by a single percent sign (%), and %c is replaced by a colon (:).

       SYSTEM
              If $SYSTEM is set, it will have the same effect as if it had been specified as the argument to the -m option.

       MANOPT
              If  $MANOPT is set, it will be parsed prior to man's command line and is expected to be in a similar format.  As all of the other man specific environment variables can be expressed as command line
              options, and are thus candidates for being included in $MANOPT it is expected that they will become obsolete.  N.B.  All spaces that should be interpreted as part of an option's  argument  must  be
              escaped.

       MANWIDTH
              If $MANWIDTH is set, its value is used as the line length for which manual pages should be formatted.  If it is not set, manual pages will be formatted with a line length appropriate to the current
              terminal  (using  the value of $COLUMNS, and ioctl(2) if available, or falling back to 80 characters if neither is available).  Cat pages will only be saved when the default formatting can be used,
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              Normally, when output is not being directed to a terminal (such as to a file or a pipe), formatting characters are discarded to make it easier to read the result without special tools.  However, if
              $MAN_KEEP_FORMATTING is set to any non-empty value, these formatting characters are retained.  This may be useful for wrappers around man that can interpret formatting characters.

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              with the pager's display.  Programs such as groff often produce relatively minor error messages about typographical problems such as poor alignment, which are unsightly and generally confusing when
              displayed along with the manual page.  However, some users want to see them anyway, so, if $MAN_KEEP_STDERR is set to any non-empty value, error output will be displayed as usual.

       MAN_DISABLE_SECCOMP
              On Linux, man normally confines subprocesses that handle untrusted data using a seccomp(2) sandbox.  This makes it safer to run complex parsing code over arbitrary manual pages.  If this goes wrong
              for some reason unrelated to the content of the page being displayed, you can set $MAN_DISABLE_SECCOMP to any non-empty value to disable the sandbox.

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              If the $PIPELINE_DEBUG environment variable is set to "1", then man will print debugging messages to standard error describing each subprocess it runs.

       LANG, LC_MESSAGES
              Depending  on  system and implementation, either or both of $LANG and $LC_MESSAGES will be interrogated for the current message locale.  man will display its messages in that locale (if available).
              See setlocale(3) for precise details.

FILES
       /etc/manpath.config
              man-db configuration file.

       /usr/share/man
              A global manual page hierarchy.

STANDARDS
       POSIX.1-2001, POSIX.1-2008, POSIX.1-2017.

SEE ALSO
       apropos(1), groff(1), less(1), manpath(1), nroff(1), troff(1), whatis(1), zsoelim(1), manpath(5), man(7), catman(8), mandb(8)

       Documentation for some packages may be available in other formats, such as info(1) or HTML.

HISTORY
 Manual page man(1) line 382/458 97% (press h for help or q to quit)
```
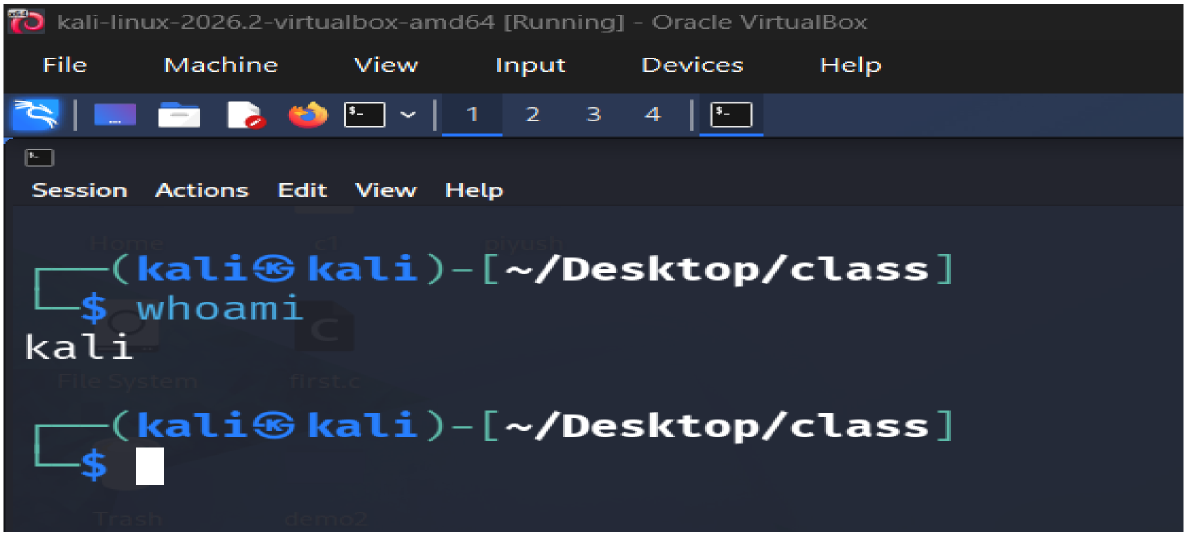
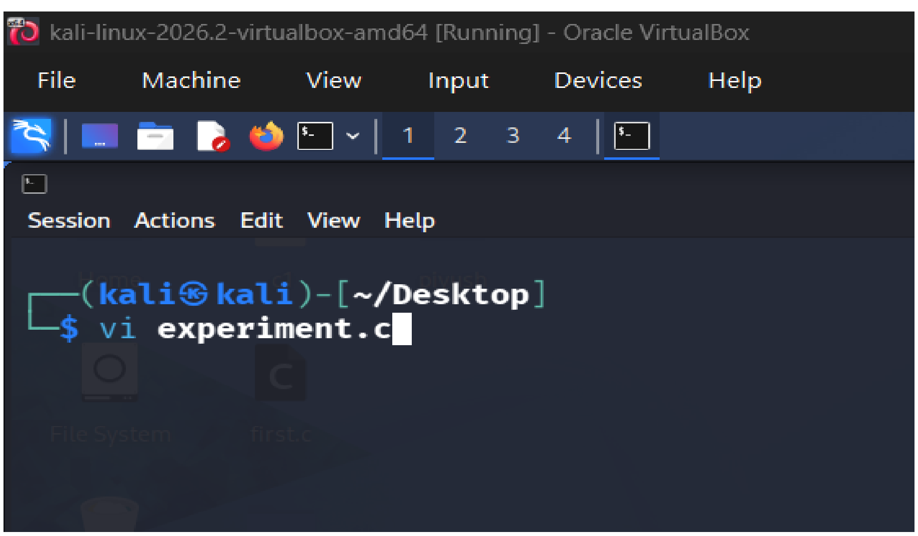

11) whoami - Displays the username of the currently logged-in user.

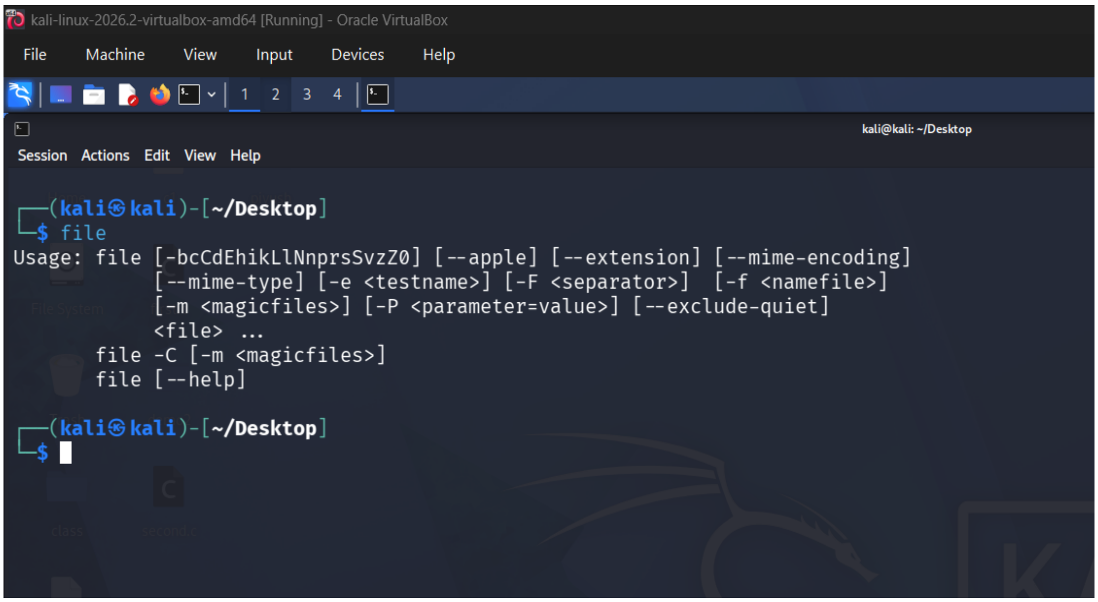


- **File Command**

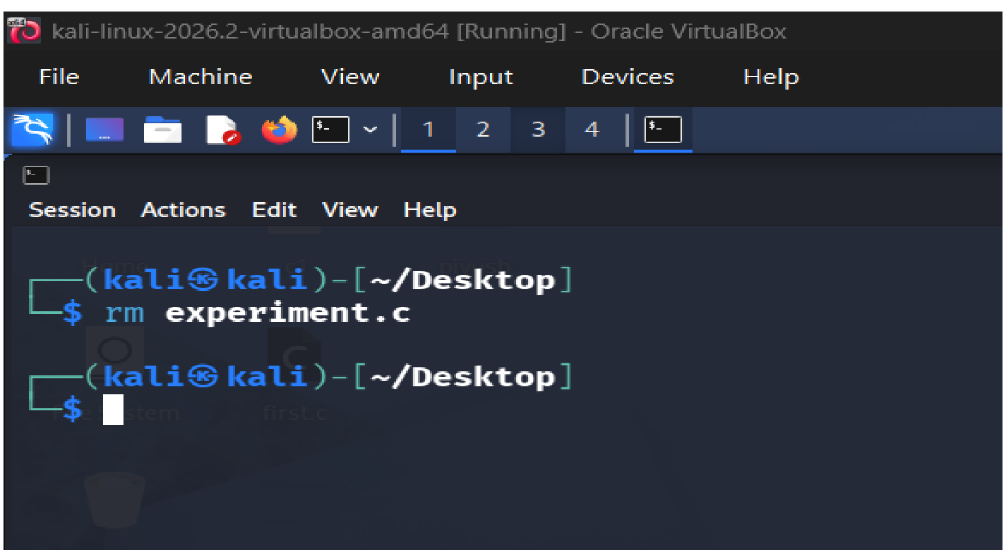
1) vi - Opens the Vi editor to create or edit a file.



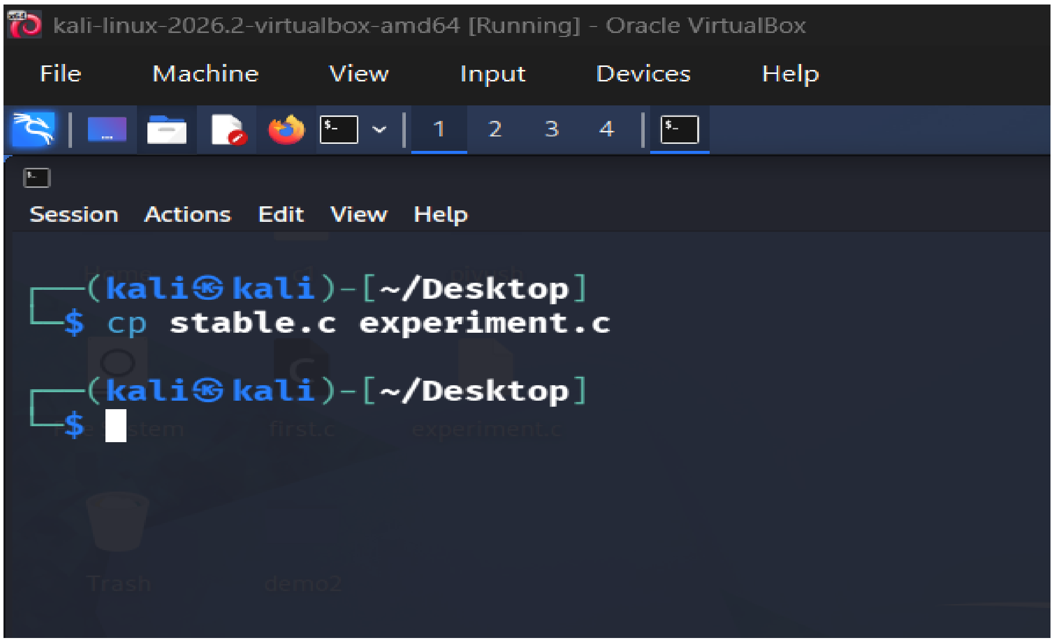
2) file - Displays information about the type of a file.



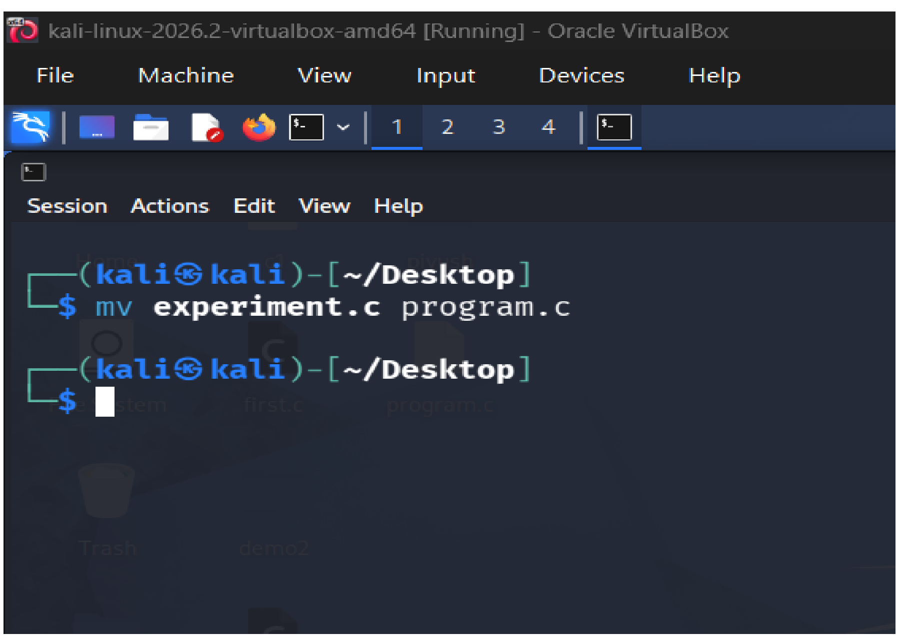
3) rm - Removes/deletes a file.



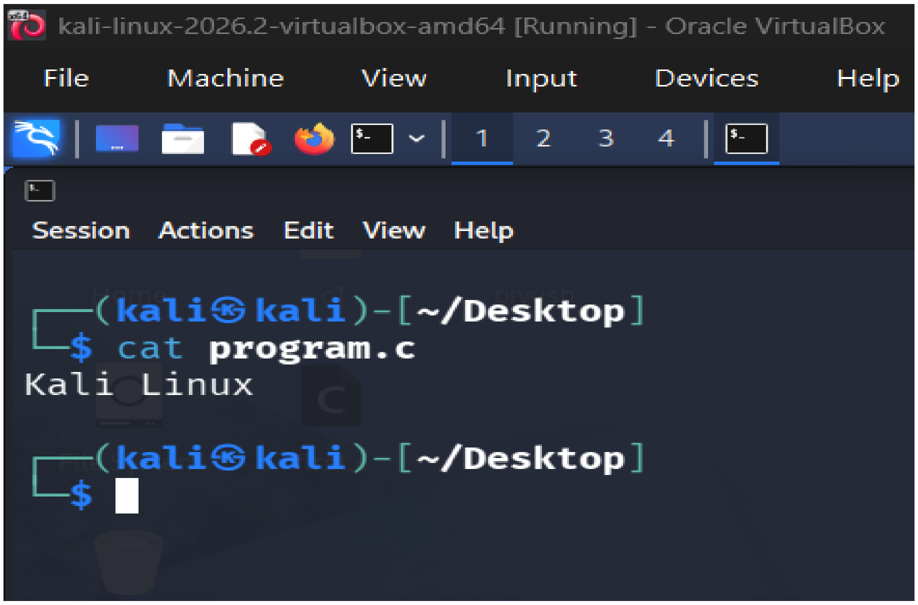
4) cp - Copies a file from one location to another.



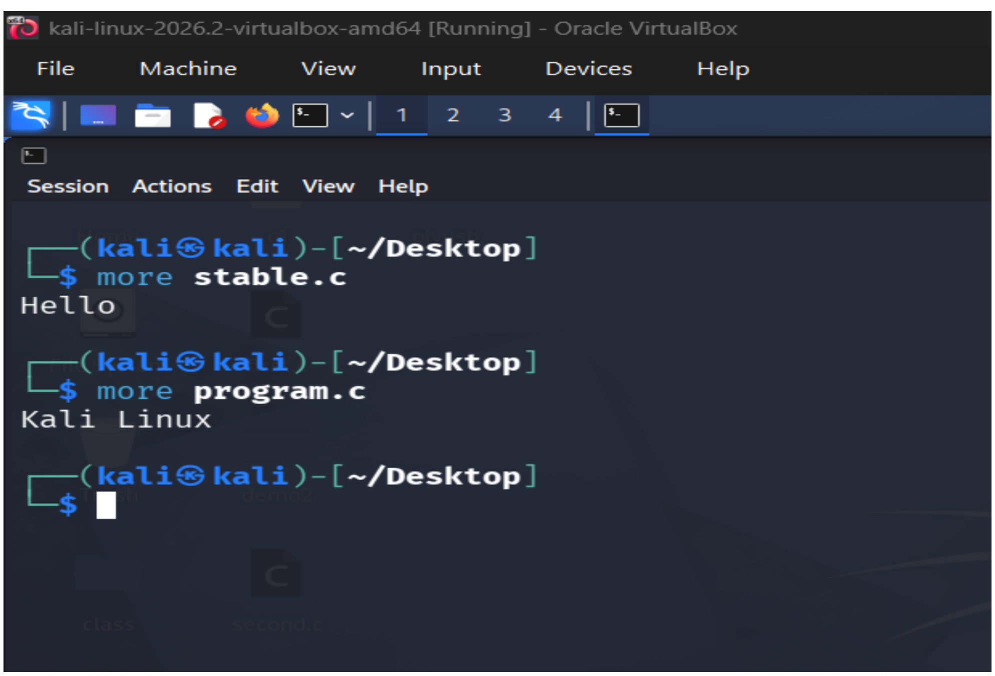
5) mv - Moves or renames a file.



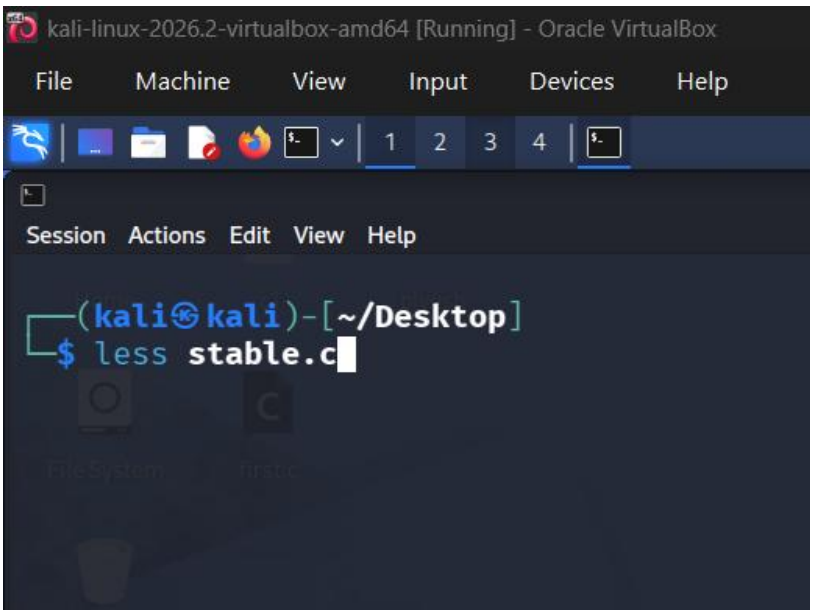
6) cat - Displays the contents of a file.



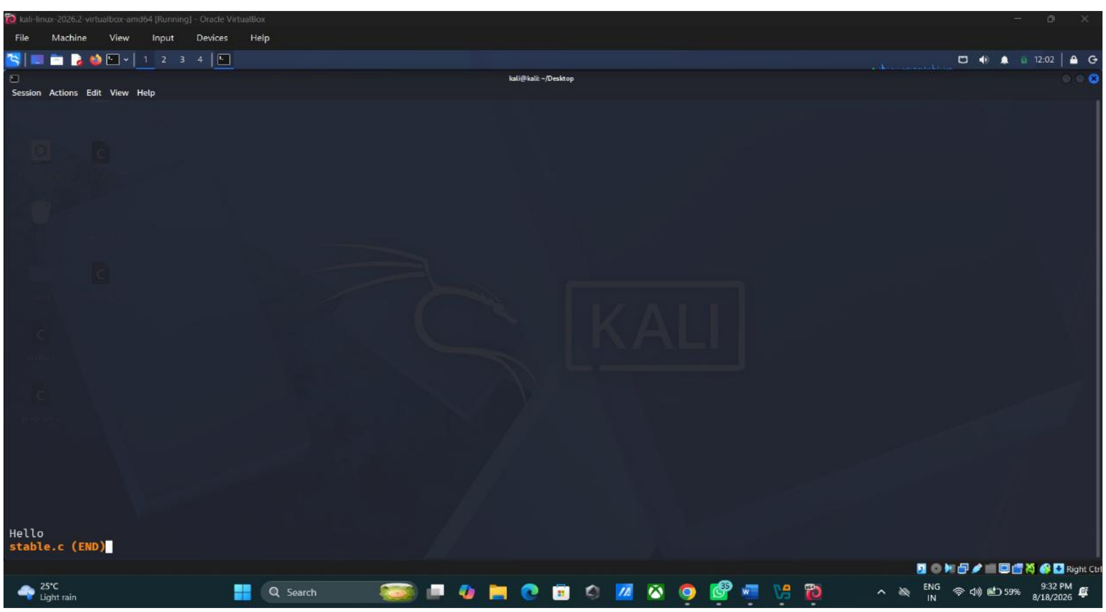
7) more - Displays file contents one screen at a time.



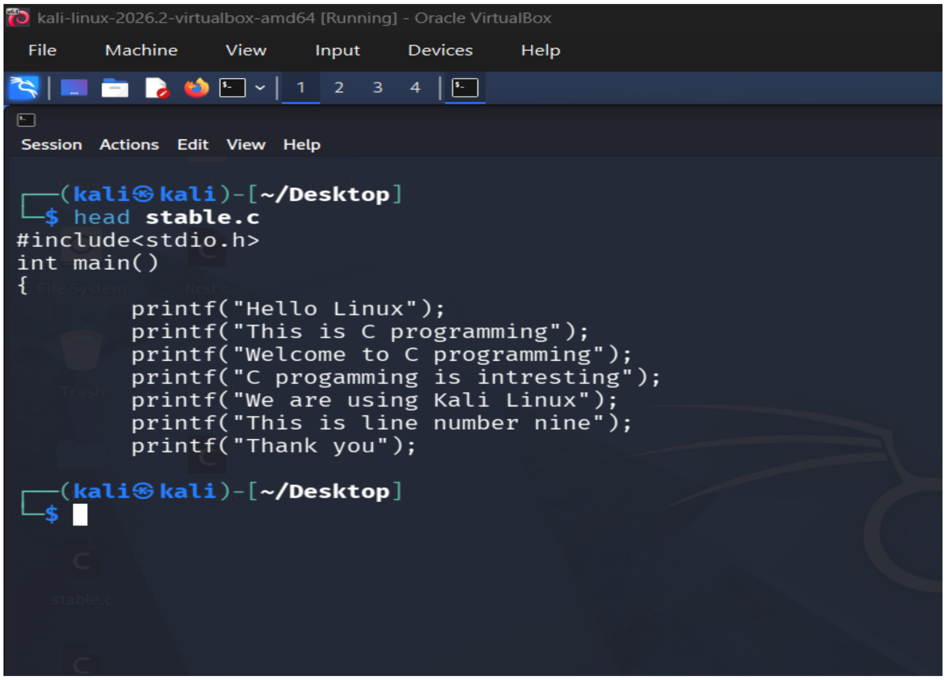
8) less - Displays file contents interactively and allows scrolling.



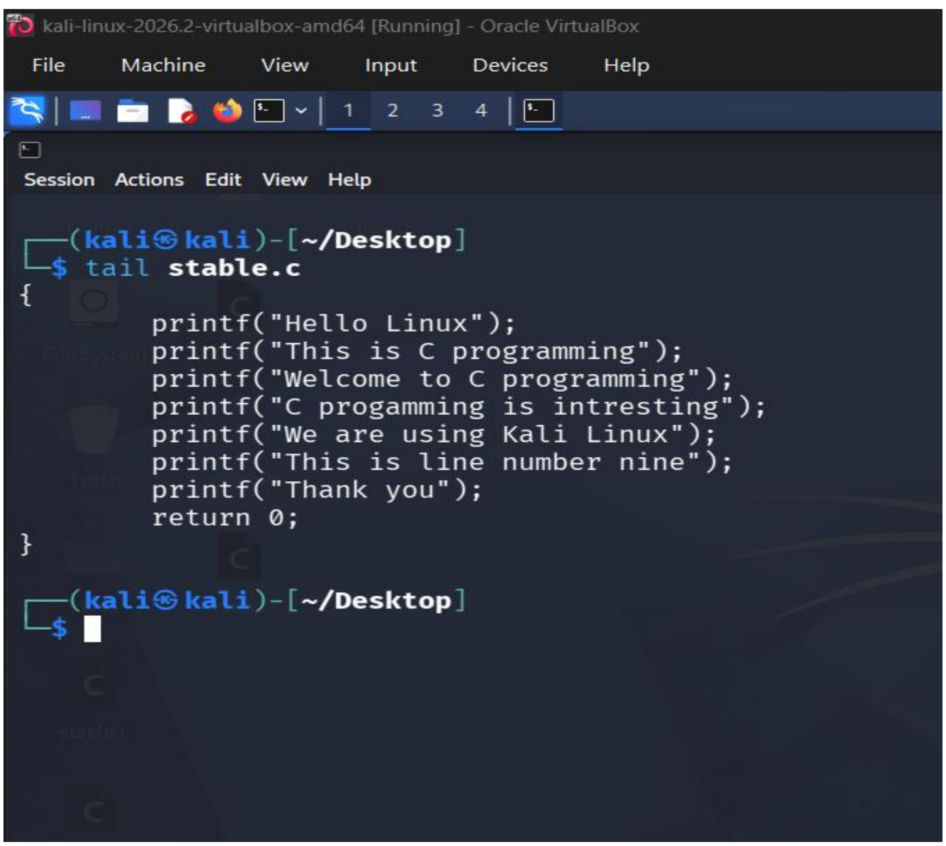
After executing the less command.



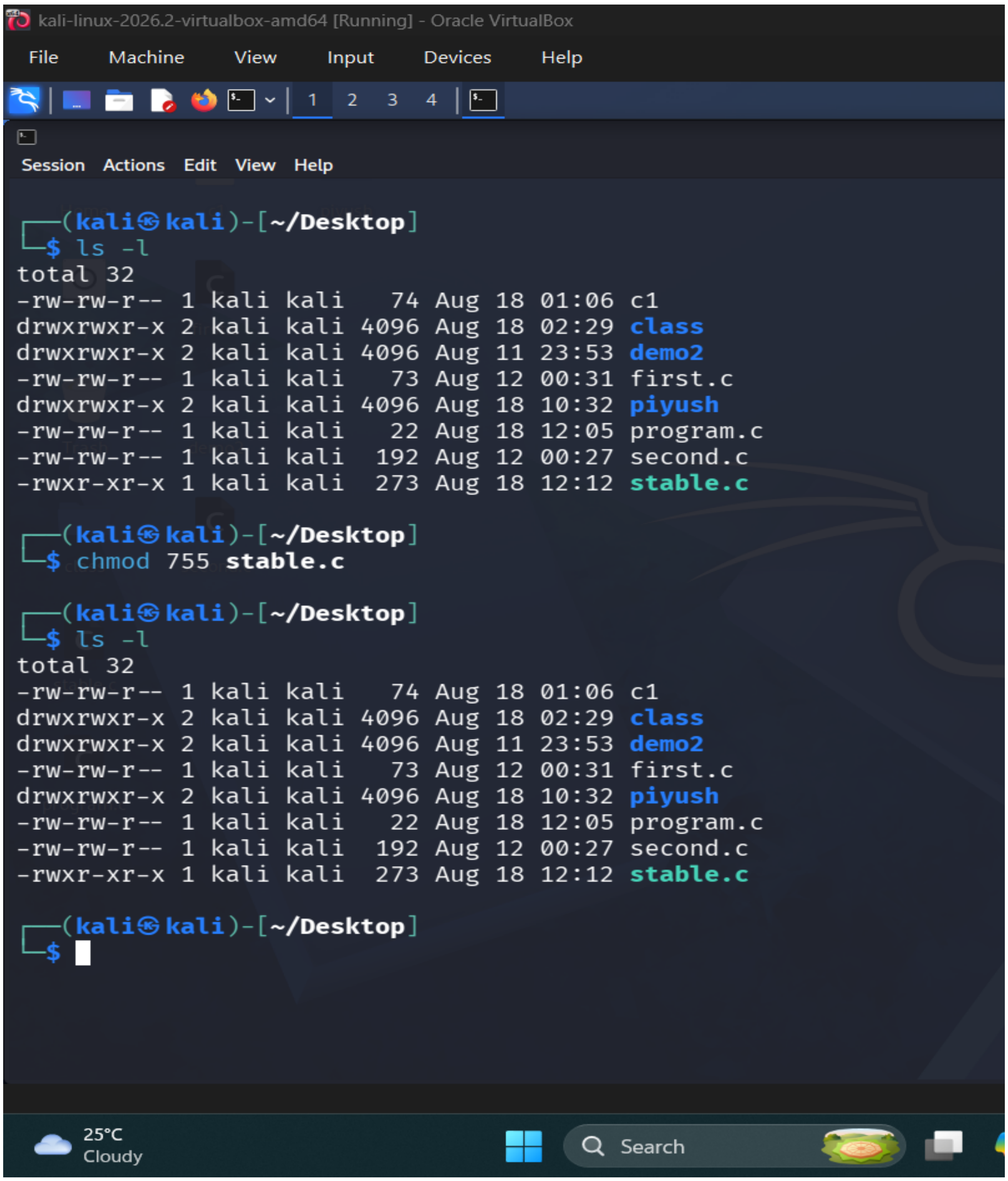
9) head – Displays the first few lines of a file.



10) tail - Displays the last few lines of a file.

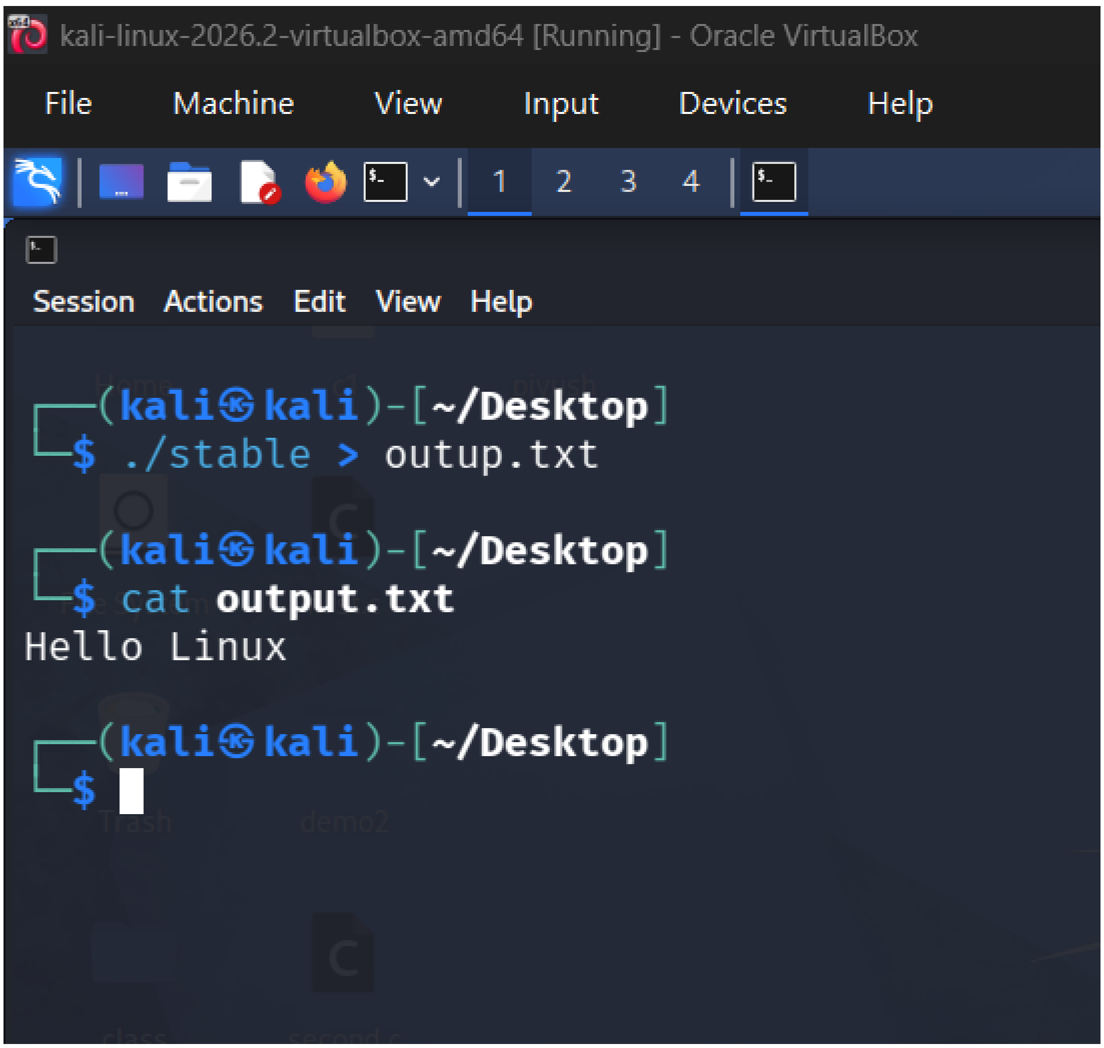


11) chmod - Changes the permissions of a file or directory.

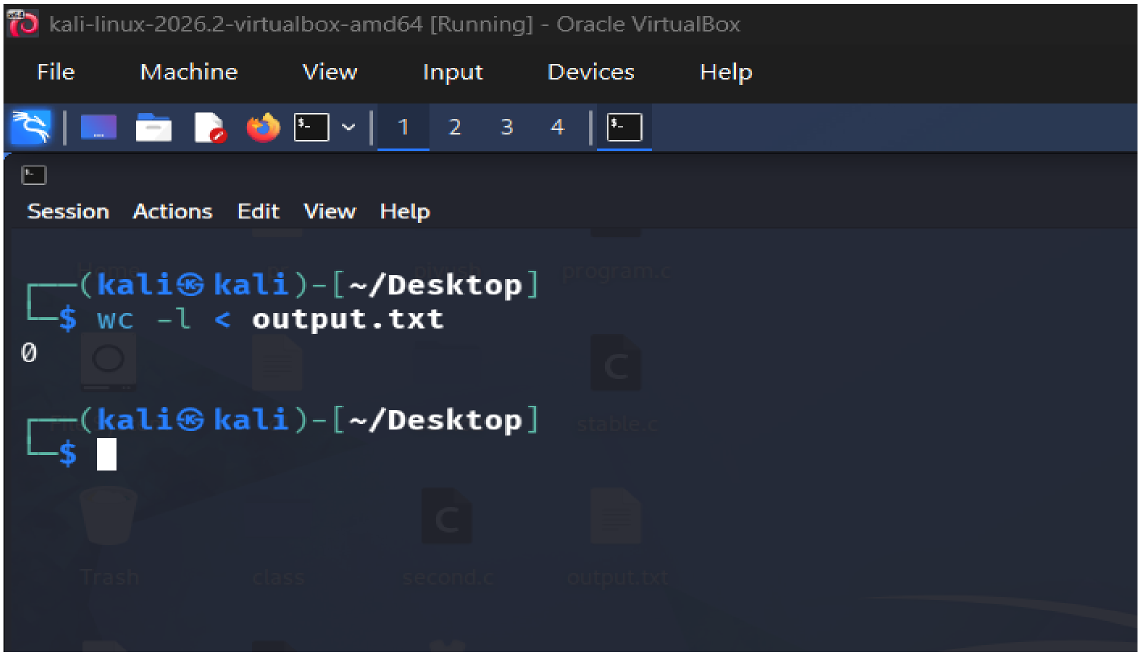


- Redirections command

1) > - This operator sends the output of a command to a file. If the file already exists, its contents are overwritten.



2) < - This operators takes input from a file instead of the keyboard.



3) >> - This operators adds the output to the end of an existing file without overwriting its previous contents.